MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, $J\overline{K}$, DIGITAL, KEY CODE QHK

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic $J\overline{K}$ flip-flop module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0879 failures per million operating hours for class I or 0.1196 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be QHK.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/329-1 for class I and M28787/329-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

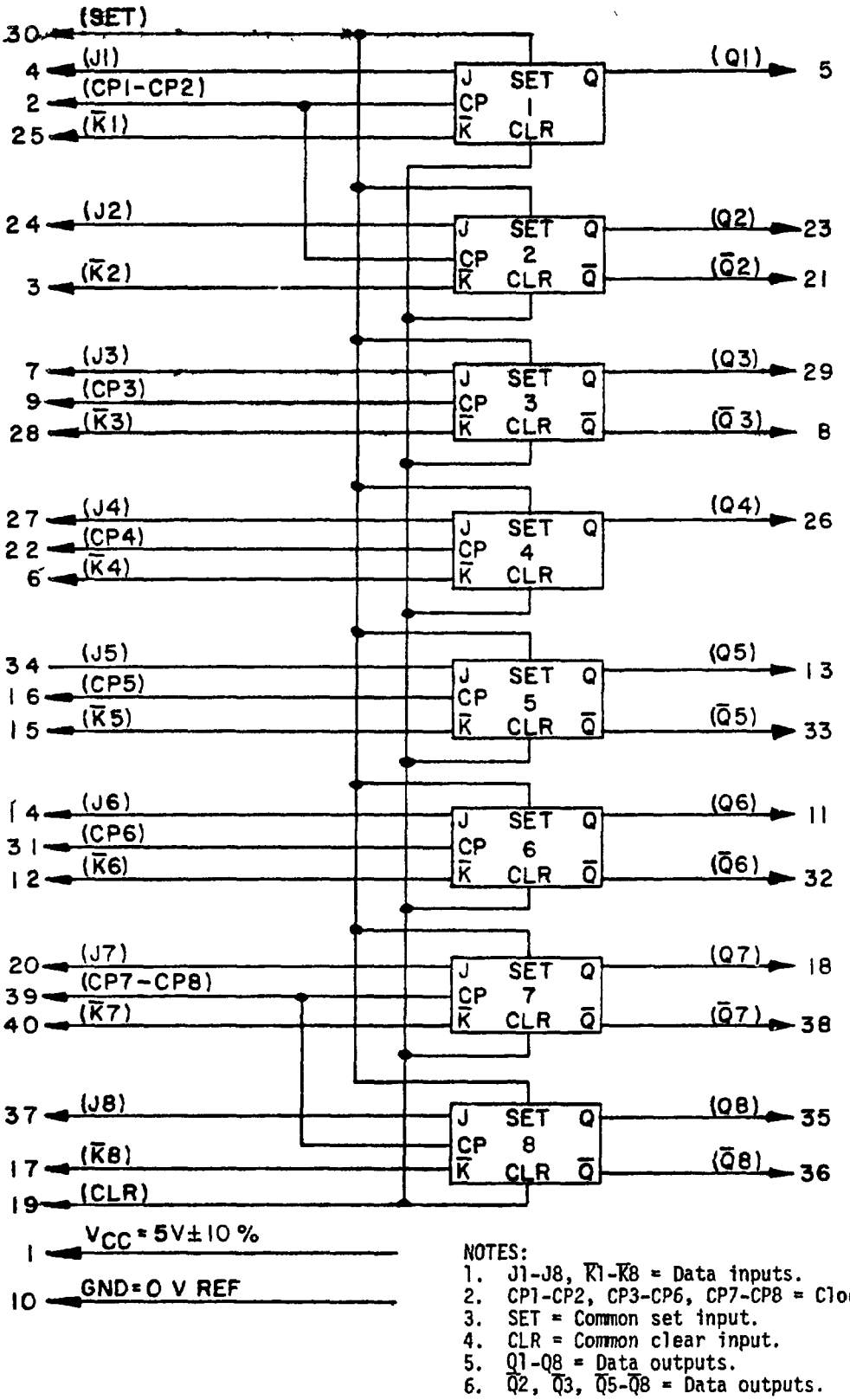
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. J1-J8, K̄1-K̄8 = Data inputs.
2. CP1-CP2, CP3-CP6, CP7-CP8 = Clock inputs
3. SET = Common set input.
4. CLR = Common clear input.
5. Q1-Q8 = Data outputs.
6. Q̄2, Q̄3, Q̄5-Q̄8 = Data outputs.

FIGURE 1. QHK module function diagram, JK̄ flip-flop.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{CC} stress	IV	S	← Module must pass all remaining tests →										
C1,C2, C24-C26	Input breakdown current	V	S	0	2	0	2	0	2	0	2	0	2	mA
C3-C22, C27-C30	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
C23	Input breakdown current	V	S	0	4	0	4	0	4	0	4	0	4	mA
D1-D14	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	-40	-100	mA
E1-E14	Output leakage current	VII	X	0	1	0	1	0	1	0	1	0	1	mA
F1-F24	Input clamp diode voltage	VIII	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G8	Functional	IX	X	← Output must be as specified in table IX →										
H1-H14	Output voltage, low level	X	X	0	500	0	500	0	500	0	500	0	500	mV
J1-J14	Output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K3, K25,K26	Input current, high level	XI	X	0	200	0	200	0	200	0	200	0	200	µA
K4-K8, K27-K30	Input current, high level	XI	X	0	100	0	100	0	100	0	100	0	100	µA
K9-K24,	Input current, high level	XI	X	0	50	0	50	0	50	0	50	0	50	µA
L1-L3	Input current, low level	XII	X	-0.1	-8.0	-0.1	-8.0	-0.1	-8.0	-0.1	-8.0	-0.1	-8.0	mA
L4	Input current, low level	XII	X	-0.1	-4.0	-0.1	-4.0	-0.1	-4.0	-0.1	-4.0	-0.1	-4.0	mA
L5-L8	Input current, low level	XII	X	-0.10	-4.32	-0.10	-4.32	-0.10	-4.32	-0.10	-4.32	-0.10	-4.32	mA
L9-L24	Input current, low level	XII	X	-0.10	-2.16	-0.10	-2.16	-0.10	-2.16	-0.10	-2.16	-0.10	-2.16	mA

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M	Power supply current	XIII	S	0.1	262.0	0.1	262.0	0.1	262.0	0.1	262.0	0.1	262.0	mA
N1-N14	Propagation delay time from clock (t_{PLH}), C_L = 50 pF	XIV	S	3	13	3	13	3	15	3	13	3	17	ns
P1-P14	Propagation delay time from clock (t_{PHL}), C_L = 50 pF	XIV	S	4	15	4	15	4	17	4	15	4	19	ns
Q1-Q8	Propagation delay time from set (t_{PLH}), C_L = 50 pF	XV	S	4	14	4	14	4	16	4	14	4	18	ns
Q9-Q16	Propagation delay time from set (t_{PLH}), C_L = 50 pF	XV	Q	4	14	4	14	4	16	4	14	4	18	ns
R1-R6	Propagation delay time from set (t_{PHL}), C_L = 50 pF	XV	S	6	21	6	21	6	23	6	21	6	25	ns
R7-R12	Propagation delay time from set (t_{PHL}), C_L = 50 pF	XV	Q	6	21	6	21	6	23	6	21	6	25	ns
S1-S6	Propagation delay time from clear (t_{PLH}), C_L = 50 pF	XV	S	4	14	4	14	4	16	4	14	4	18	ns
S7-S12	Propagation delay time from clear (t_{PLH}), C_L = 50 pF	XV	Q	4	14	4	14	4	16	4	14	4	18	ns
T1-T8	Propagation delay time from clear (t_{PHL}), C_L = 50 pF	XV	S	6	21	6	21	6	23	6	21	6	25	ns
T9-T16	Propagation delay time from clear (t_{PHL}), C_L = 50 pF	XV	Q	6	21	6	21	6	23	6	21	6	25	ns
U1-U14	Operating frequency and minimum clock pulse width check, C_L = 50 pF	XVI	S	90	110	90	110	90	110	90	110	90	110	ns
U15-U28	Operating frequency and minimum clock pulse width check, C_L = 50 pF	XVI	Q	20	30	20	30	20	30	20	30	20	30	ns
V1-V14	Output crosstalk, low level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
W1-W14	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
X	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
Y	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
Z1	Input capacitance	XX	Q	5	60	5	60	5	60	6	60	5	60	pF
Z2-Z4	Input capacitance	XX	Q	5	30	5	30	5	30	6	30	5	30	pF
Z5-Z24	Input capacitance	XX	Q	5	15	5	15	5	15	6	15	5	15	pF
AA1-AA28	Setup time	XXI	S	80	120	80	120	80	120	80	120	80	120	ns
BB1-BB28	Hold time	XXII	S	80	120	80	120	80	120	80	120	80	120	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through BB in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through Y, AA, and BB shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Y, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through Y, AA, and BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through BB shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life parameters from table I.

	Class I	Class II	Unit
Power dissipation	1.44 max	1.44 max	W
Input voltage			
Low level	-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High level	2 min, V_{CC} max	2 min, V_{CC} max	V

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	Class I	Class II	Unit
Input current, CLR			
Low level (V_{IL} = 0.0 V)	-8 max	-8 max	mA
High level (V_{IH} = 2.7 V)	200 max	200 max	µA
(V_{IH} = 5.5 V)	4 max	4 max	mA
Input current, CP1, 2,7,8			
Low level (V_{IL} = 0.0 V)	-8 max	-8 max	mA
High level (V_{IH} = 2.7 V)	200 max	200 max	µA
(V_{IH} = 5.5 V)	2 max	2 max	mA
Input current, CP3-6			
Low level (V_{IL} = 0.0 V)	-4.32 max	-4.32 max	mA
High level (V_{IH} = 2.7 V)	100 max	100 max	µA
(V_{IH} = 5.5 V)	1 max	1 max	mA
Input current, SET			
Low level (V_{IL} = 0.0 V)	-4 max	-4 max	mA
High level (V_{IH} = 2.7 V)	100 max	100 max	µA
(V_{IH} = 5.5 V)	2 max	2 max	mA
Input current, $J\overline{K}$			
Low level (V_{IL} = 0.0 V)	-2.16 max	-2.16 max	mA
High level (V_{IH} = 2.7 V)	50 max	50 max	µA
(V_{IH} = 5.5 V)	1 max	1 max	mA
Output drive current			
Low level (V_{OL} = 0.5 V)	20 max	20 max	mA
High level (V_{OH} = 2.5 V)	-1 max	-1 max	µA
Operating frequency	75 max	75 max	MHz
Setup time			
t_{setup} (J, $\overline{K}$)	6 min	6 min	ns
Hold time			
t_{hold}(J, $\overline{K}$)	0 min	0 min	ns
Pulse width			
t_{PW} (CLOCK)	8 min	8 min	ns
t_{PW} (SET, CLR)	8 min	8 min	ns

6.5 Module function. This module contains eight $J\overline{K}$ FLIP-FLOPS with common buffered asynchronous SET and CLEAR inputs (active high) for all FLIP-FLOPS. The clock input is common to FLIP-FLOPS 1 and 2 (CP1-CP2) and to FLIP-FLOPS 7 and 8 (CP7-CP8). FLIP-FLOPS 3-6 each have an independent clock input. FLIP-FLOPS 2, 3, 5, 6, 7, and 8 have both a Q and $\overline{Q}$ output. FLIP-FLOPS 1 and 4 have only a Q output. Information on the inputs is transferred to the output on the positive clock input transition.

6.5.1 Input to output relationship. Each $J\overline{K}$ FLIP-FLOP has the input to output relationship of table II for positive logic.

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TABLE II. Input to output relationship. 1/ - 5/

Inputs					Outputs	
SET	CLEAR	CLOCK	J	$\overline{K}$	Q	$\overline{Q}$
H	L	X	X	X	H	L
L	H	X	X	X	L	H
H	H	X	X	X	H	H
					6/	6/
L	L	↑	L	L	L	H
L	L	↑	H	L	Toggle	
L	L	↑	L	H	Q_o	$\overline{Q}_o$
L	L	↑	H	H	H	L
L	L	L	X	X	Q_o	$\overline{Q}_o$

1/ H = High level (steady state).
2/ L = Low level (steady state).
3/ X = Irrelevant (any input, including transitions).
4/ ↑ = Transition from low to high.
5/ Q_o, $\overline{Q}_o$ = The level of Q, $\overline{Q}$ respectively, before the indicated steady-state input conditions were established.
6/ This configuration is unstable; that is, it will not persist when SET and CLEAR return to their inactive (low-level) state.

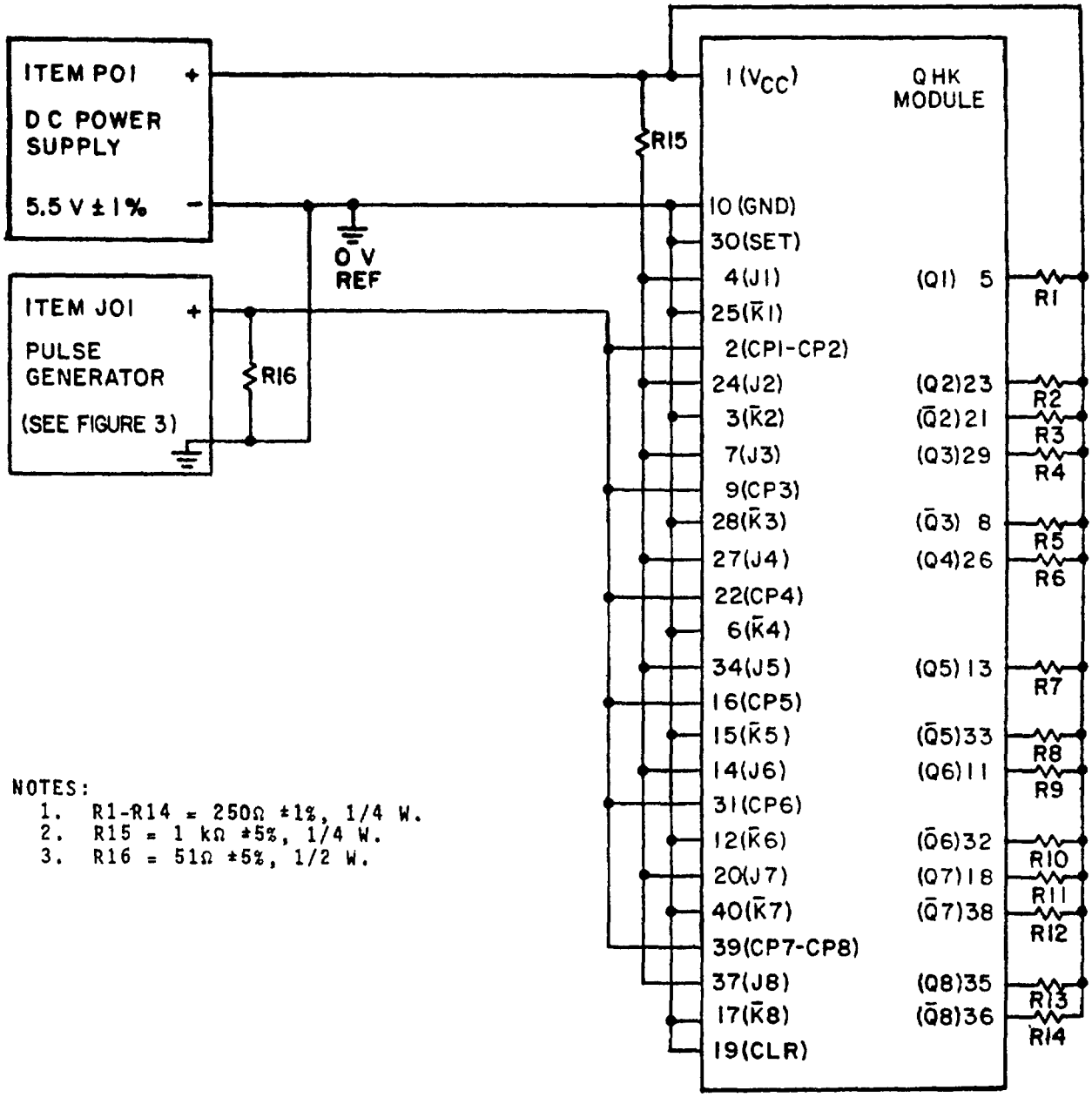
6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

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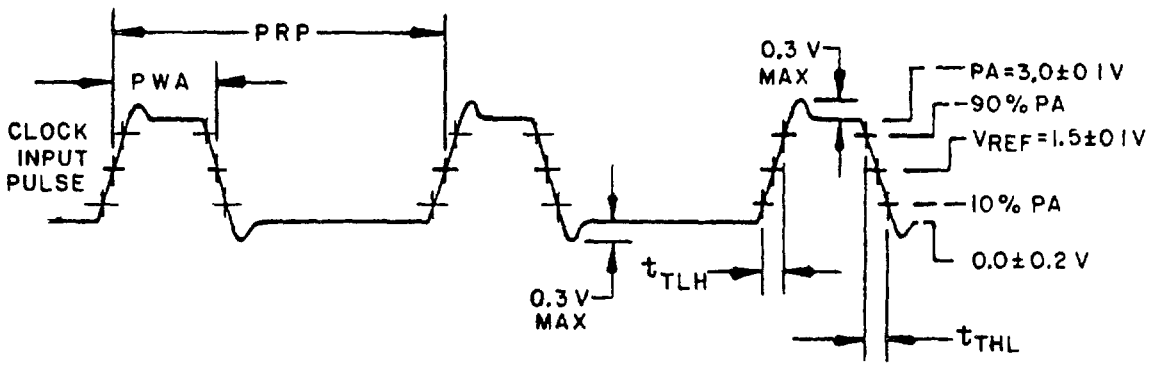
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NOTES:
1. R1-R14 = 250Ω ±1%, 1/4 W.
2. R15 = 1 kΩ ±5%, 1/4 W.
3. R16 = 51Ω ±5%, 1/2 W.

FIGURE 2. Operating temperature and life test circuit.

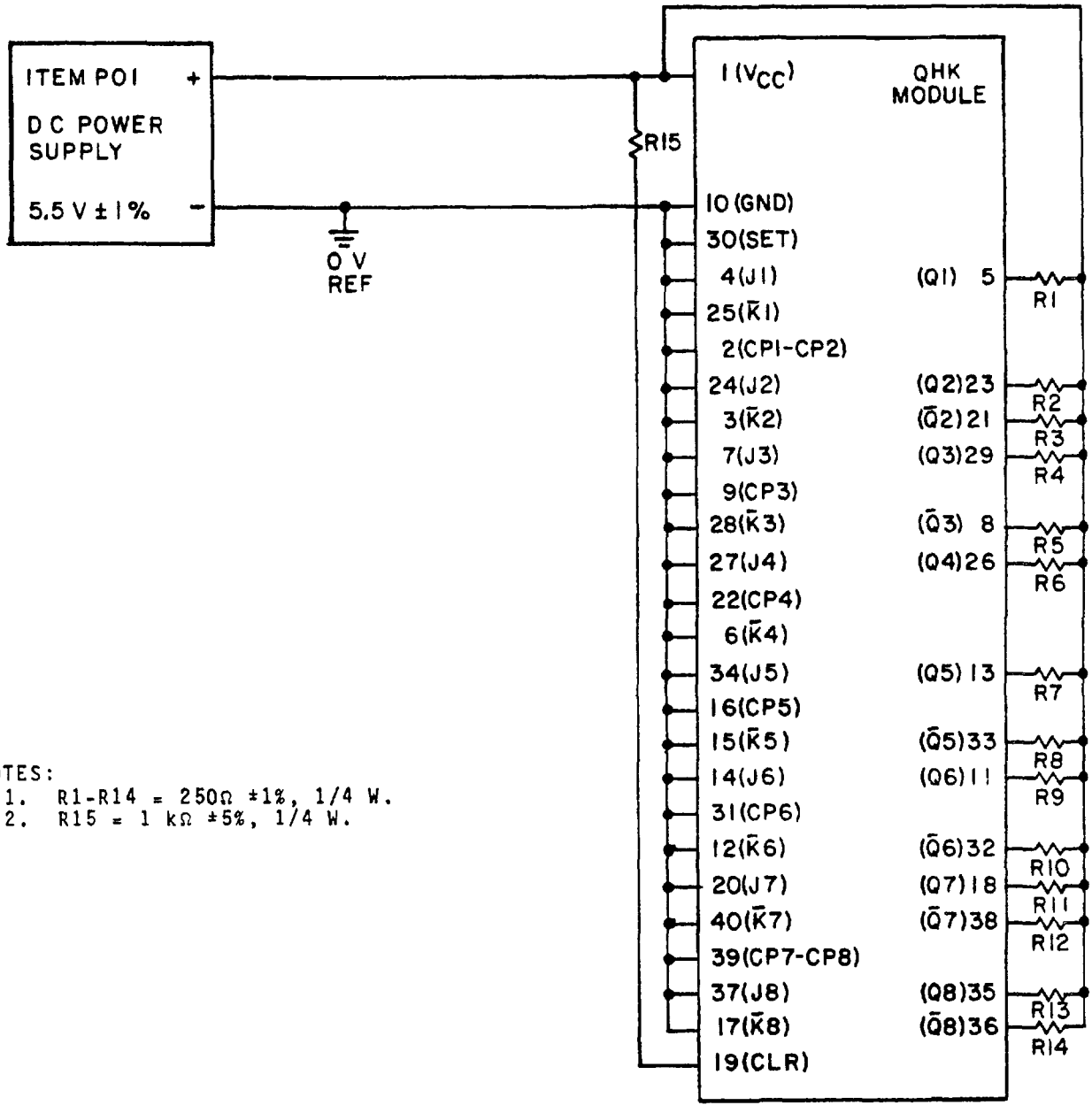
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NOTES:
1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns.
   PWA = 250 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltages are referenced to ground.

FIGURE 3. Operating temperature and life test circuit waveform.

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NOTES:
1. R1-R14 = 250Ω ±1%, 1/4 W.
2. R15 = 1 kΩ ±5%, 1/4 W.

FIGURE 4. Component temperature test circuit.

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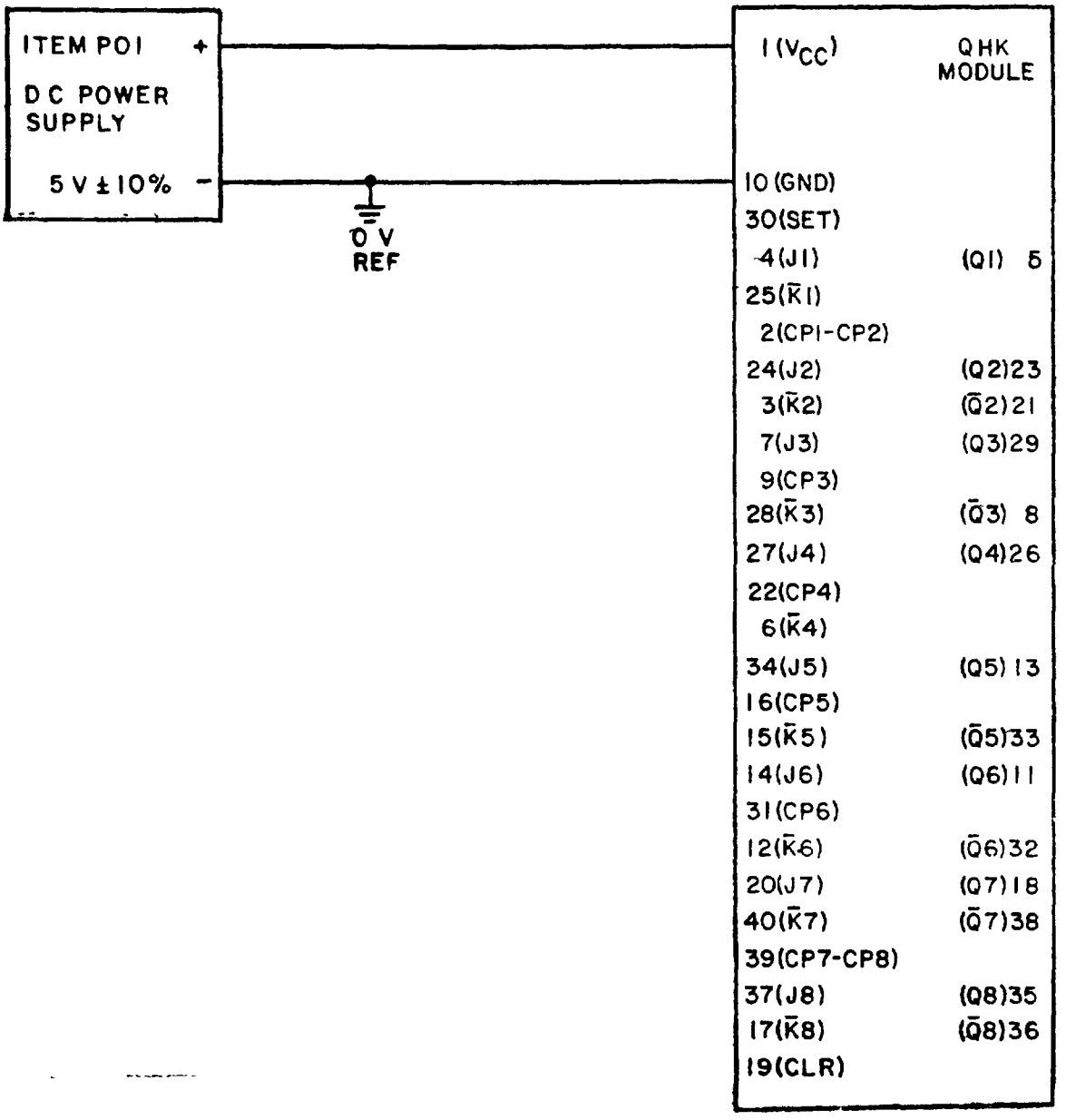


FIGURE 5. Vibration test circuit.

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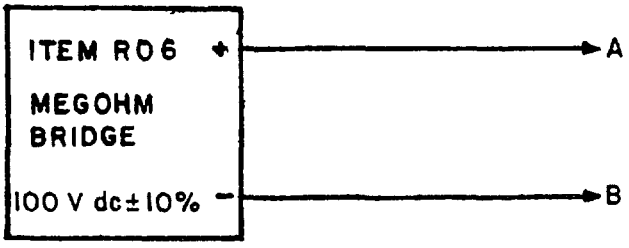
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	6

TABLE IV. V_{CC} stress test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
B	V_{CC} stress	← Module must pass remaining tests →				7

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1 (V_{CC})	QHK MODULE
10 (GND)	
30 (SET)	
4 (J1)	(Q1) 5
25 ($\bar{K}$1)	
2 (CP1-CP2)	
24 (J2)	(Q2) 23
3 ($\bar{K}$2)	($\bar{Q}$2) 21
7 (J3)	(Q3) 29
9 (CP3)	
28 ($\bar{K}$3)	($\bar{Q}$3) 8
27 (J4)	(Q4) 26
22 (CP4)	
6 ($\bar{K}$4)	
34 (J5)	(Q5) 13
16 (CP5)	
15 ($\bar{K}$5)	($\bar{Q}$5) 33
14 (J6)	(Q6) 11
31 (CP6)	
12 ($\bar{K}$6)	($\bar{Q}$6) 32
20 (J7)	(Q7) 18
40 ($\bar{K}$7)	($\bar{Q}$7) 38
39 (CP7-CP8)	
37 (J8)	(Q8) 35
17 ($\bar{K}$8)	($\bar{Q}$8) 36
19 (CLR)	

WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Isolation test circuit for table III.

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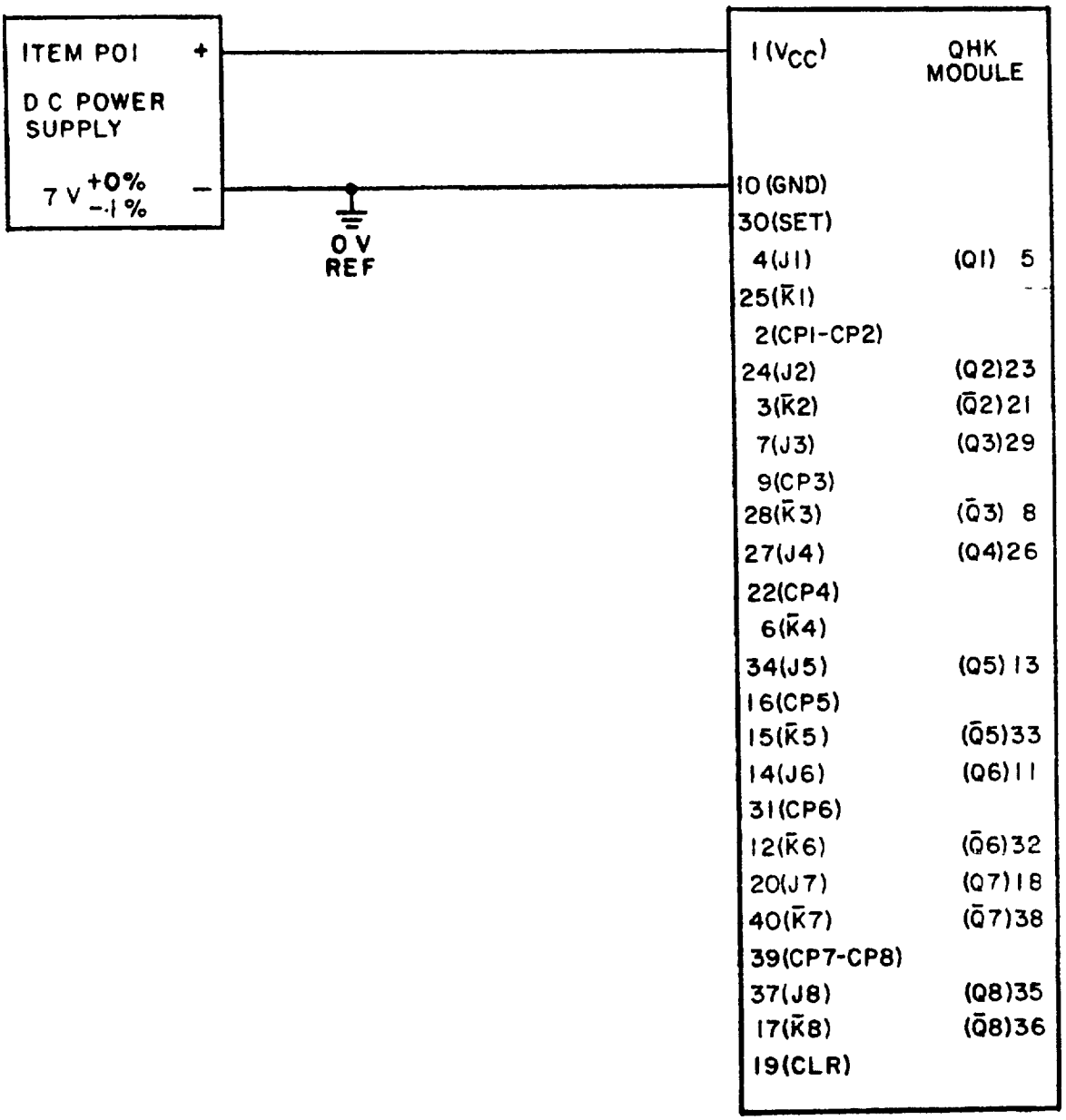


FIGURE 7. V_{CC} stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	CP1-CP2	Input breakdown current	3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	2	0	2	mA	8
C2	CP7-CP8	"	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,40	39	"	2	"	"
C3	CP3	"	2,3,4,6,7,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	9	"	1	"	"
C4	CP4	"	2,3,4,6,7,9,12,14,15,16,17, 20,24,25,27,28,30,31, 34,37,39,40	22	"	"	"	"
C5	CP5	"	2,3,4,6,7,9,12,14,15,17, 20,22,24,25,27,28,30,31, 34,37,39,40	16	"	"	"	"
C6	CP6	"	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30, 34,37,39,40	31	"	"	"	"
C7	J1	"	2,3,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	4	"	"	"	"
C8	J2	"	2,3,4,6,7,9,12,14,15,16,17, 20,22,25,27,28,30,31, 34,37,39,40	24	"	"	"	"
C9	J3	"	2,3,4,6,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	7	"	"	"	"
C10	J4	"	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,28,30,31, 34,37,39,40	27	"	"	"	"
C11	J5	"	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 37,39,40	34	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

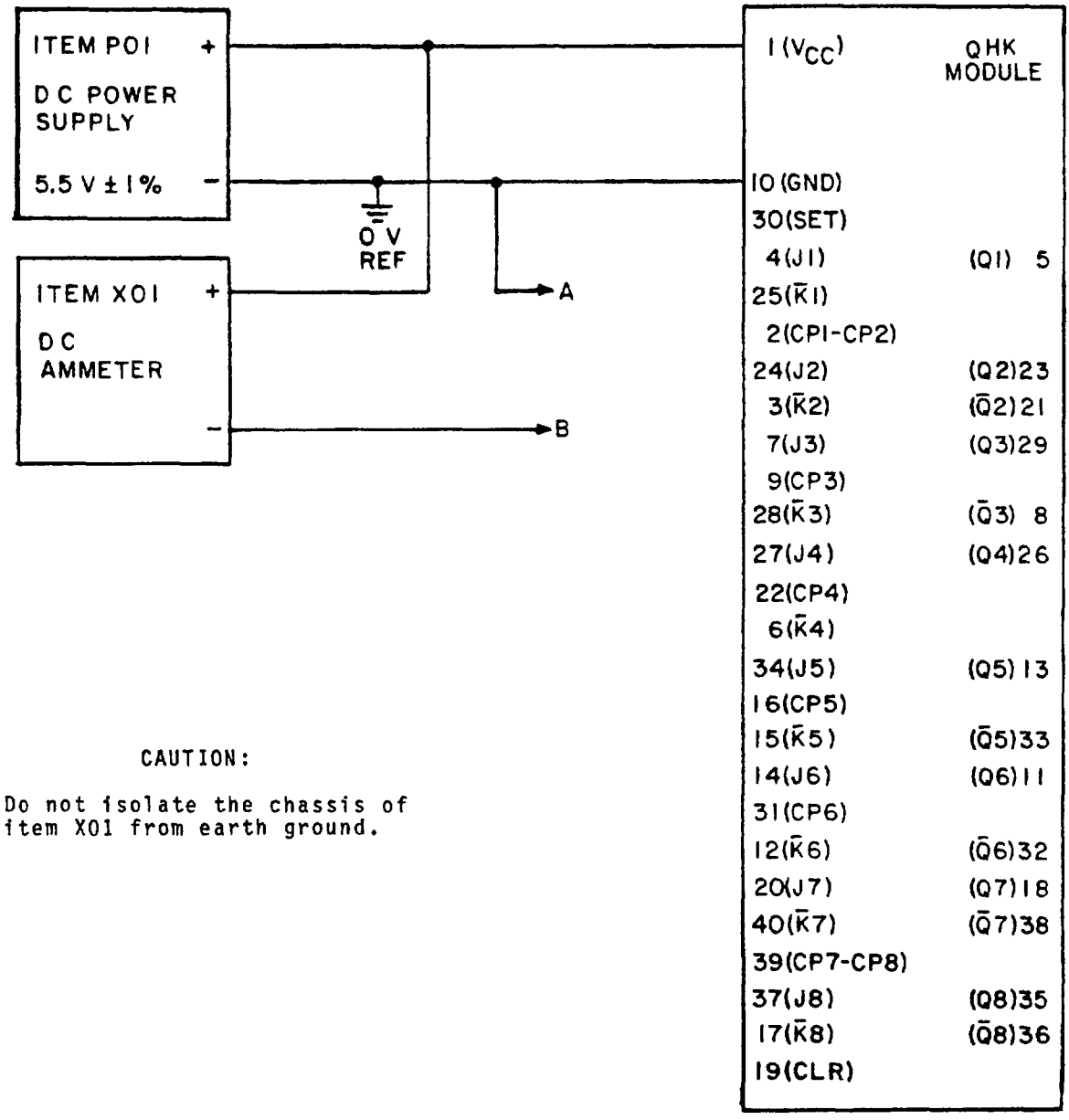
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C12	J6	Input breakdown current	2,3,4,6,7,9,12,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	14	0	1	mA	8
C13	J7	"	2,3,4,6,7,9,12,14,15,16,17, 22,24,25,27,28,30,31, 34,37,39,40	20	"	"	"	"
C14	J8	"	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,39,40	37	"	"	"	"
C15	$\overline{K}$1	"	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,27,28,30,31, 34,37,39,40	25	"	"	"	"
C16	$\overline{K}$2	"	2,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	3	"	"	"	"
C17	$\overline{K}$3	"	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,30,31, 34,37,39,40	28	"	"	"	"
C18	$\overline{K}$4	"	2,3,4,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	6	"	"	"	"
C19	$\overline{K}$5	"	2,3,4,6,7,9,12,14,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	15	"	"	"	"
C20	$\overline{K}$6	"	2,3,4,6,7,9,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	12	"	"	"	"
C21	$\overline{K}$7	"	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39	40	"	"	"	"
C22	$\overline{K}$8	"	2,3,4,6,7,9,12,14,15,16, 20,22,24,25,27,28,30,31, 34,37,39,40	17	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C23	CLR	Input breakdown current	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	19	0	4	mA	8
C24	SET	"	2,3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31, 34,37,39,40	30	"	2	"	"
C25	CP1-CP2	"	3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31, 34,37,39,40	2	"	"	"	"
C26	CP7-CP8	"	2,3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31, 34,37,40	39	"	"	"	"
C27	CP3	"	2,3,4,6,7,12,14,15,16,17, 19,20,22,24,25,27,28,31, 34,37,39,40	9	"	1	"	"
C28	CP4	"	2,3,4,6,7,9,12,14,15,16,17, 19,20,24,25,27,28,31, 34,37,39,40	22	"	"	"	"
C29	CP5	"	2,3,4,6,7,9,12,14,15,17, 19,20,22,24,25,27,28,31, 34,37,39,40	16	"	"	"	"
C30	CP6	"	2,3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28, 34,37,39,40	31	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

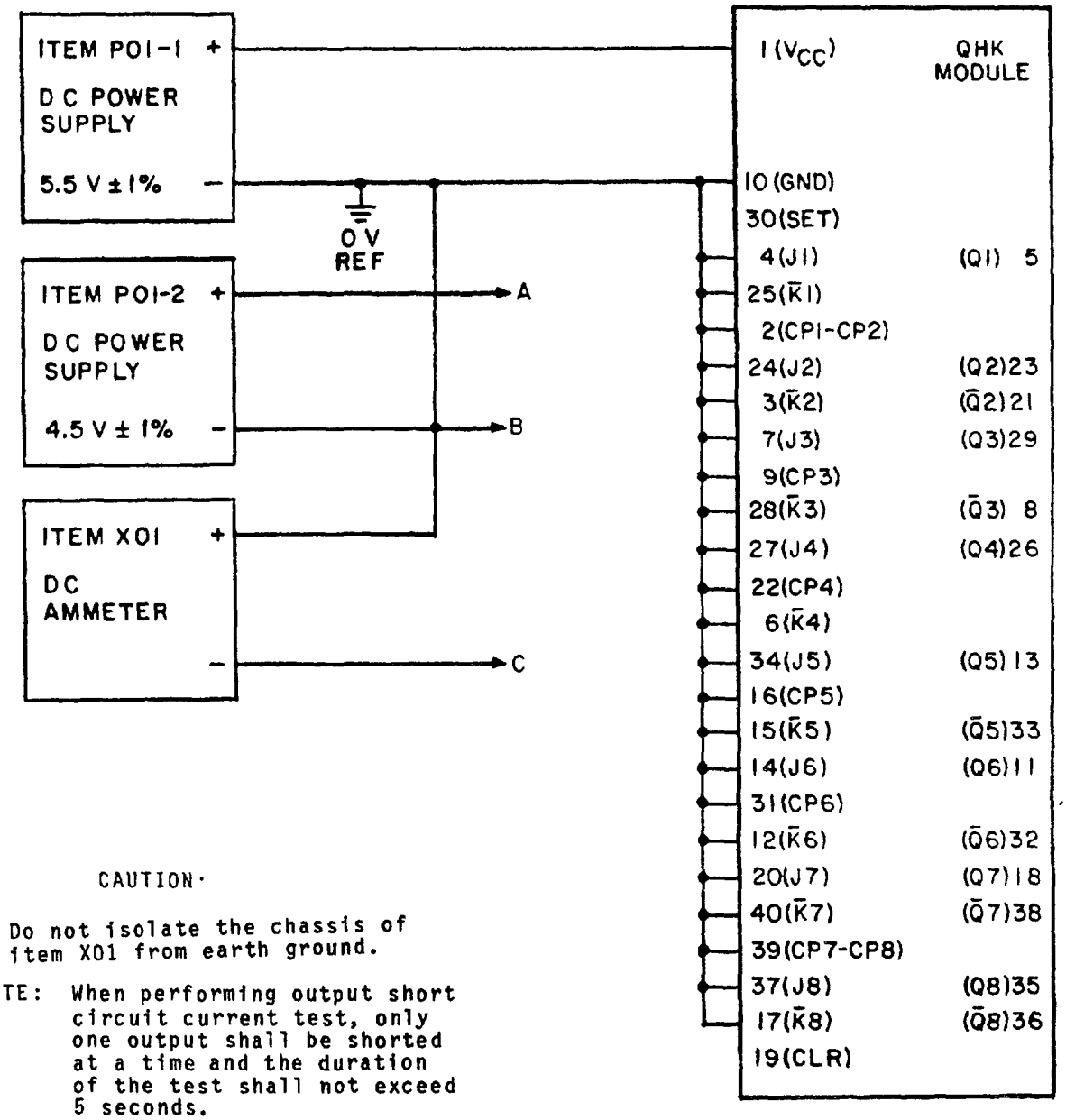
FIGURE 8. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	Q1 Output short circuit current	30	19	5	-40	-100	mA	9
D2	Q2 "	"	"	23	"	"	"	"
D3	Q3 "	"	"	29	"	"	"	"
D4	Q4 "	"	"	26	"	"	"	"
D5	Q5 "	"	"	13	"	"	"	"
D6	Q6 "	"	"	11	"	"	"	"
D7	Q7 "	"	"	18	"	"	"	"
D8	Q8 "	"	"	35	"	"	"	"
D9	$\overline{Q2}$ "	19	30	21	"	"	"	"
D10	$\overline{Q3}$ "	"	"	8	"	"	"	"
D11	$\overline{Q5}$ "	"	"	33	"	"	"	"
D12	$\overline{Q6}$ "	"	"	32	"	"	"	"
D13	$\overline{Q7}$ "	"	"	38	"	"	"	"
D14	$\overline{Q8}$ "	"	"	36	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: When performing output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.

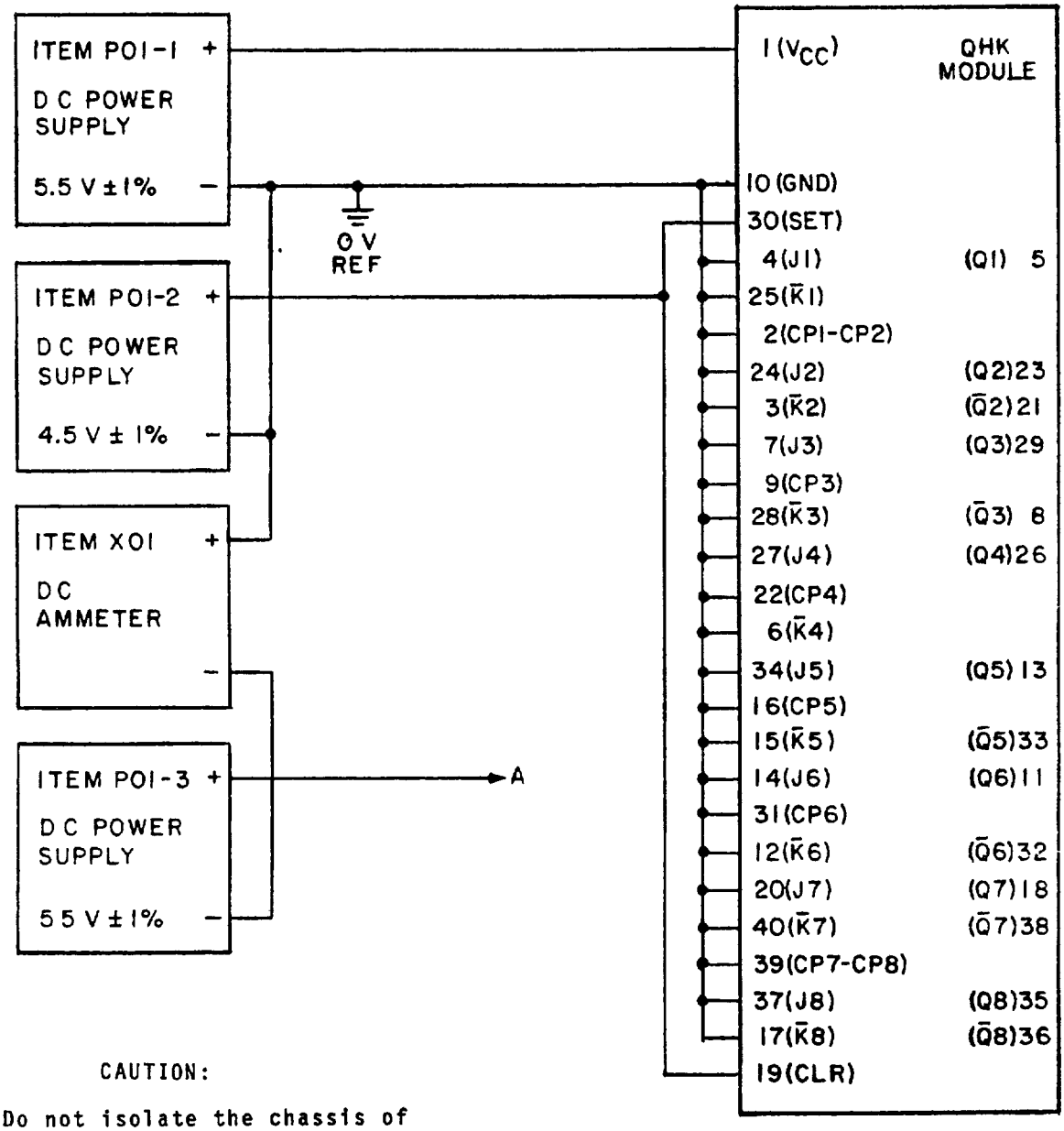
FIGURE 9. Output short circuit current test circuit for table VI.

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TABLE VII. Output leakage current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	Q1 Output leakage current	5	0	1	mA	10
E2	Q2 "	23	"	"	"	"
E3	$\overline{Q2}$ "	21	"	"	"	"
E4	Q3 "	29	"	"	"	"
E5	$\overline{Q3}$ "	8	"	"	"	"
E6	Q4 "	26	"	"	"	"
E7	Q5 "	13	"	"	"	"
E8	$\overline{Q5}$ "	33	"	"	"	"
E9	Q6 "	11	"	"	"	"
E10	$\overline{Q6}$ "	32	"	"	"	"
E11	Q7 "	18	"	"	"	"
E12	$\overline{Q7}$ "	38	"	"	"	"
E13	Q8 "	35	"	"	"	"
E14	$\overline{Q8}$ "	36	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: Item PO1-3 shall be current limited to 5 mA.

FIGURE 10. Output leakage current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

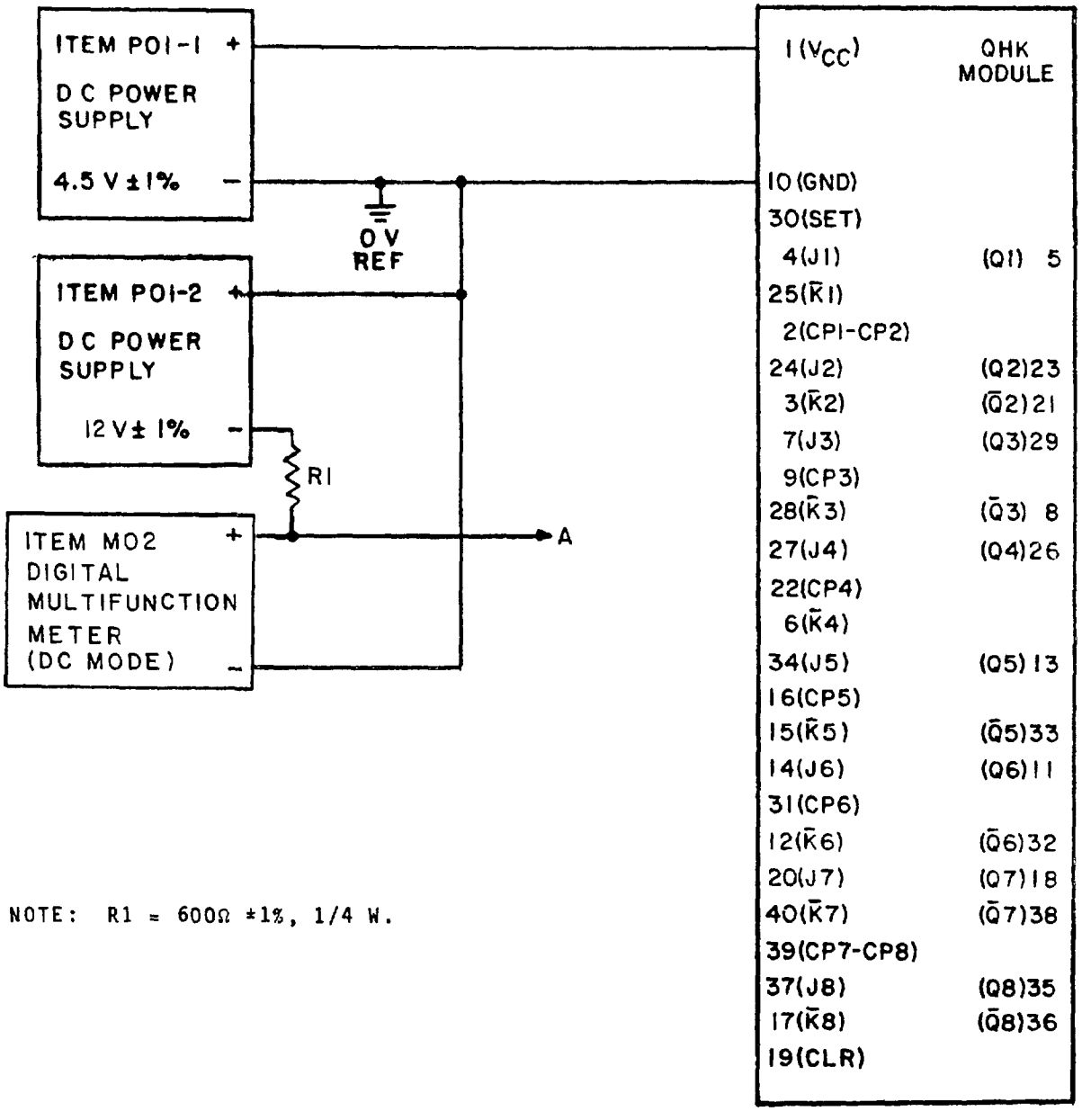
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F1	SET Input clamp diode voltage	30	-0.4	-1.2	V	11
F2	CLR "	19	"	"	"	"
F3	CP1-CP2 "	2	"	"	"	"
F4	CP3 "	9	"	"	"	"
F5	CP4 "	22	"	"	"	"
F6	CP5 "	16	"	"	"	"
F7	CP6 "	31	"	"	"	"
F8	CP7-CP8 "	39	"	"	"	"
F9	J1 "	4	"	"	"	"
F10	J2 "	24	"	"	"	"
F11	J3 "	7	"	"	"	"
F12	J4 "	27	"	"	"	"
F13	J5 "	34	"	"	"	"
F14	J6 "	14	"	"	"	"
F15	J7 "	20	"	"	"	"
F16	J8 "	37	"	"	"	"
F17	$\overline{K1}$ "	25	"	"	"	"
F18	$\overline{K2}$ "	3	"	"	"	"
F19	$\overline{K3}$ "	28	"	"	"	"
F20	$\overline{K4}$ "	6	"	"	"	"

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TABLE VIII. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
F21	$\overline{K5}$	Input clamp diode voltage	15	-0.4	-1.2	V	11
F22	$\overline{K6}$	"	12	"	"	"	"
F23	$\overline{K7}$	"	40	"	"	"	"
F24	$\overline{K8}$	"	17	"	"	"	"

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NOTE: R1 = 600Ω ±1%, 1/4 W.

FIGURE 11. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Functional test.

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	$\overline{Q2}$	$\overline{Q3}$	$\overline{Q5}$	$\overline{Q6}$	$\overline{Q7}$	$\overline{Q8}$	Min	Max		
G1	Functional	19,4,25,24,3,7, 28,27,6,34,15, 14,12,20,40,37, 17,	30	2/ 3/	2,9, 22,16, 31,39	H	H	H	H	H	H	H	H	L	L	L	L	L	L	1/			12,13
G2	"	30	19,4,25,24,3,7, 28,27,6,34,15, 14,12,20,40,37, 17	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	L	"			"
G3	"	Open	30,19,4,25,24, 7,28,27,6,34, 15,14,12,20,40, 37,17,3	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
G4	"	30,19	4,25,24,3,7,28, 27,6,34,15,14, 12,20,40,37,17	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
G5	"	30,19,4,25,24, 3,7,28,27,6,34, 15,14,12,20,40, 37,17	Open	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
G6	"	30,19,4,24,7, 27,34,14,20,37	25,3,28,6,15, 12,40,17	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
G7	"	30,19,25,3,28, 6,15,12,40,17	4,24,7,27,34, 14,20,37	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
G8	"	30,19,4,24,7, 27,34,14,20,37	25,3,28,6,15, 12,40,17	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"

1/ Output logic states
   a. Low level 0 to 500 mV.
   b. High level 2.5 to 4 5 V
2/ For test conditions G1-G8, verification of the output states in footnote 1 is obtained by connecting item M02 (point C) to each output pin in succession after input conditions have been accomplished. Test conditions G1-G8 are to be performed in sequence
3/ Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, $\overline{Q2}$, $\overline{Q3}$, $\overline{Q5}$, $\overline{Q6}$, $\overline{Q7}$, $\overline{Q8}$ output logic states are measured at output pins, 5, 23, 29, 26, 13, 11, 18, 35, 21, 8, 33, 32, 38, 36

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TABLE X. Output voltage (low-level and high-level) and input threshold check test.

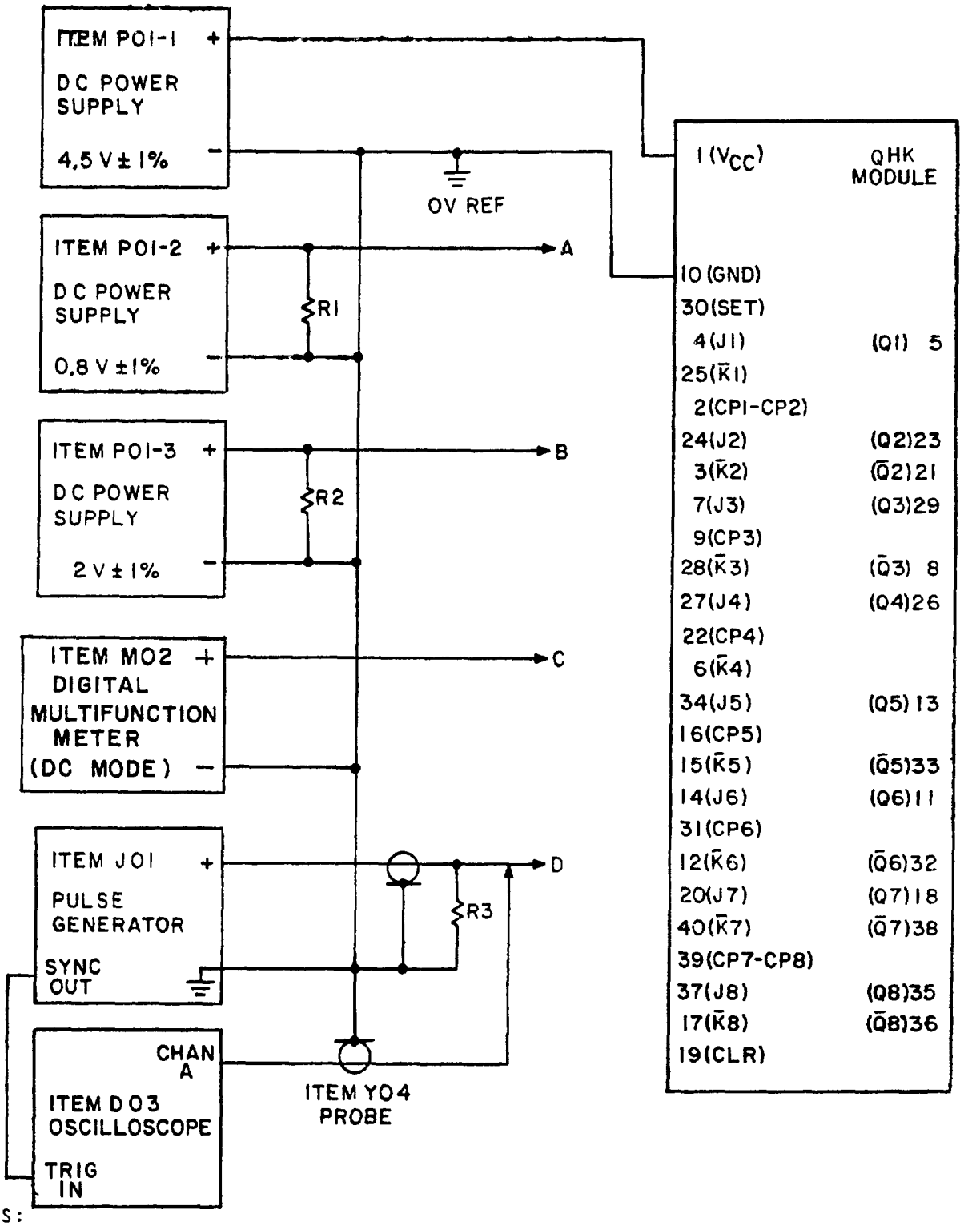
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	Q1	Output voltage, low level	30	19	5	Open	0	500	mV	12
H2	Q2	"	"	"	23	"	"	"	"	"
H3	Q3	"	"	"	29	"	"	"	"	"
H4	Q4	"	"	"	26	"	"	"	"	"
H5	Q5	"	"	"	13	"	"	"	"	"
H6	Q6	"	"	"	11	"	"	"	"	"
H7	Q7	"	"	"	18	"	"	"	"	"
H8	Q8	"	"	"	35	"	"	"	"	"
H9	$\overline{Q2}$	"	19	30	21	"	"	"	"	"
H10	$\overline{Q3}$	"	"	"	8	"	"	"	"	"
H11	$\overline{Q5}$	"	"	"	33	"	"	"	"	"
H12	$\overline{Q6}$	"	"	"	32	"	"	"	"	"
H13	$\overline{Q7}$	"	"	"	38	"	"	"	"	"
H14	$\overline{Q8}$	"	"	"	36	"	"	"	"	"
J1	Q1	Output voltage, high level	3,6,7,12,14,15, 17,19,20,24,27, 28,30,34,37,40	4,25	5	2,9,16,22, 31,39	2.5	4.5	V	12,13
J2	Q2	"	4,6,7,12,14,15, 17,19,20,25,27, 28,30,34,37,40	3,24	23	"	"	"	"	"
J3	Q3	"	3,4,6,12,14,15, 17,19,20,24,25, 27,30,34,37,40	7,28	29	"	"	"	"	"

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TABLE X. Output voltage (low-level and high-level) and input threshold check test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J4	Q4 Output voltage, high level	3,4,7,12,14,15, 17,19,20,24,25, 28,30,34,37,40	6,27	26	2,9,16,22, 31,39	2.5	4.5	V	12,13
J5	Q5 "	3,4,6,7,12,14, 17,19,20,24,25, 27,28,30,37,40	15,34	13	"	"	"	"	"
J6	Q6 "	3,4,6,7,15,17, 19,20,24,25,27, 28,30,34,37,40	12,14	11	"	"	"	"	"
J7	Q7 "	3,4,6,7,12,14, 15,17,19,24,25, 27,28,30,34,37	20,40	18	"	"	"	"	"
J8	Q8 "	3,4,6,7,12,14, 15,19,20,24,25, 27,28,30,34,40	17,37	35	"	"	"	"	"
J9	$\overline{Q2}$ "	3,19,24,30	4,6,7,12,14,15, 17,20,25,27,28, 34,37,40	21	"	"	"	"	"
J10	$\overline{Q3}$ "	7,19,28,30	3,4,6,12,14,15, 17,20,24,25,27, 34,37,40	8	"	"	"	"	"
J11	$\overline{Q5}$ "	15,19,30,34	3,4,6,7,12,14,17, 20,24,25,27,28, 37,40	33	"	"	"	"	"
J12	$\overline{Q6}$ "	12,14,19,30	3,4,6,7,15,17, 20,24,25,27,28, 34,37,40	32	"	"	"	"	"
J13	$\overline{Q7}$ "	19,20,30,40	3,4,6,7,12,14,15, 17,24,25,27,28, 34,37	38	"	"	"	"	"
J14	$\overline{Q8}$ "	17,19,30,37	3,4,6,7,12,14,15, 20,24,25,27,28, 34,40	36	"	"	"	"	"

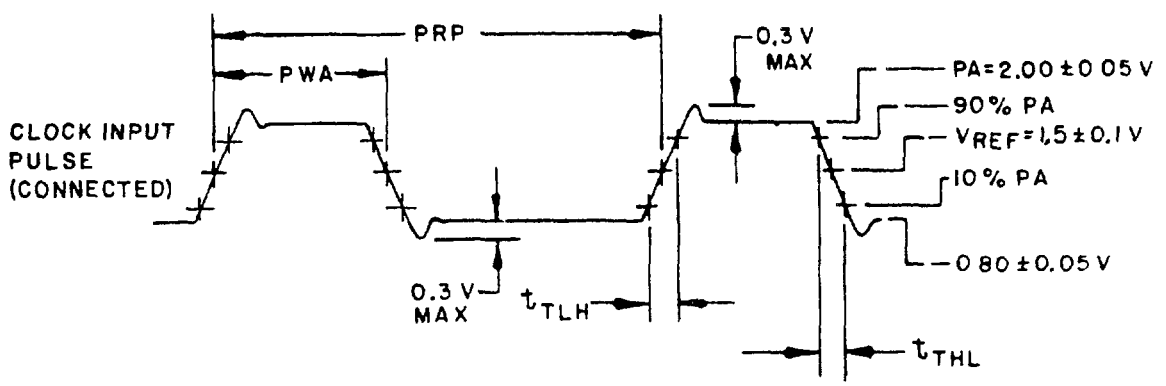
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NOTES:
1. R1 = R2 = R3 = 51Ω ±5%, 1/2 W.
2. For test conditions G1-G8 and J1-J14, make all input circuit connections for a test condition then use the single pulse mode on item J01 (after initial setup) and apply a single pulse before making each measurement.
3. For test conditions H1-H14 and J1-J14, load the output under test and the corresponding complement output (where available) with the load circuit as shown on figure 14. Load circuits are not required for test conditions G1-G8.

FIGURE 12. Functional, output voltage (low-level and high-level), and input threshold check test circuit for tables IX and X.

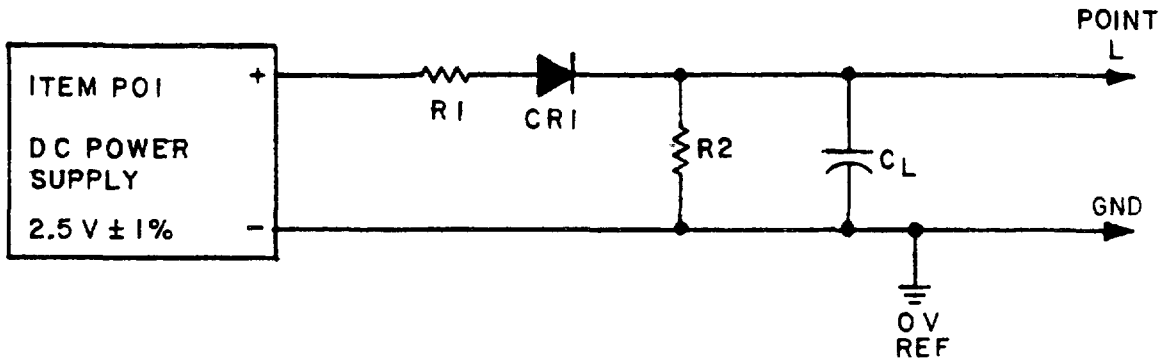
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NOTES:
1. Clock input pulse parameters:
   PWA = 500 ±100 ns.
   PRP = 1,000 ±200 ns.
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.
2. Clock threshold level setup:
   With the clock input connected, place item J01 in the single pulse mode. Use item D03 to adjust dc offset of item J01 for a low level of 0.8 ±0.5 V (clock low level threshold). Adjust item J01 to obtain the required PWA and PRP. Use item D03 to adjust item J01 amplitude for a high level of 2.00 ±0.05 V (clock high level threshold).

FIGURE 13. Functional, output voltage (high-level), and input threshold check test circuit waveform for tables IX and X.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 20 mA ±5% out of point L with point L held at 0.5 V.
3. R2 = 2.5 kΩ ±1%, 1/8 W.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance).
5. Capacitor C_L is not required for test conditions H1-H14 and J1-J14.
6. Lead length from load circuit to module output pin under test shall be less than 2 inches (50.8 mm).

FIGURE 14. QHK logic load circuit.

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TABLE XI. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	CLR Input current, high level	19	2,3,4,6,7,9,12,14,15, 16,17,20,22,24,25,27, 28,30,31,34,37,39,40	0	200	µA	15
K2	CP1-CP2 "	2	3,4,6,7,9,12,14,15, 16,17,20,22,24,25,27, 28,30,31,34,37,39,40	"	"	"	"
K3	CP7-CP8 "	39	2,3,4,6,7,9,12,14,15, 16,17,20,22,24,25,27, 28,30,31,34,37,40	"	"	"	"
K4	SET "	30	2,3,4,6,7,9,12,14,15, 16,17,19,20,22,24,25,27, 28,31,34,37,39,40	"	100	"	"
K5	CP3 "	9	2,3,4,6,7,12,14,15, 16,17,20,22,24,25,27, 28,30,31,34,37,39,40	"	"	"	"
K6	CP4 "	22	2,3,4,6,7,9,12,14,15, 16,17,20,24,25,27, 28,30,31,34,37,39,40	"	"	"	"
K7	CP5 "	16	2,3,4,6,7,9,12,14,15, 17,20,22,24,25,27, 28,30,31,34,37,39,40	"	"	"	"
K8	CP6 "	31	2,3,4,6,7,9,12,14,15, 16,17,20,22,24,25,27, 28,30,34,37,39,40	"	"	"	"
K9	J1 "	4	2,3,6,7,9,12,14,15, 16,17,20,22,24,25,27, 28,30,31,34,37,39,40	"	50	"	"
K10	J2 "	24	2,3,4,6,7,9,12,14,15, 16,17,20,22,25,27, 28,30,31,34,37,39,40	"	50	"	"

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TABLE XI. Input current (high-level) test - Continued.

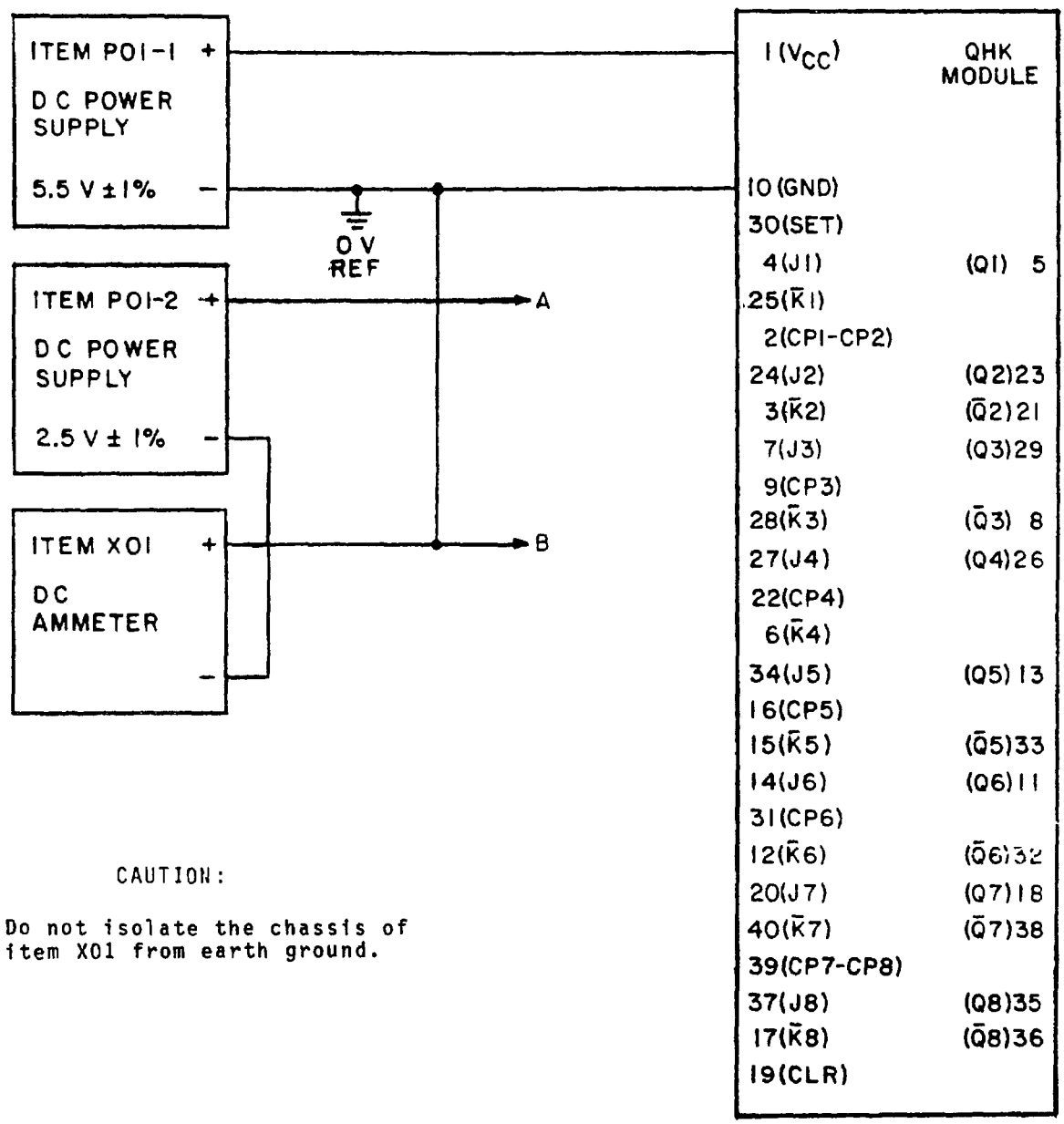
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K11	J3	Input current, high level	7	2,3,4,6,9,12,14,15, 16,17,20,22,24,25,27, 28,30,31,34,37,39,40	0	50	µA	15
K12	J4	"	27	2,3,4,6,7,9,12,14,15, 16,17,20,22,24,25, 28,30,31,34,37,39,40	"	"	"	"
K13	J5	"	34	2,3,4,6,7,9,12,14,15, 16,17,20,22,24,25,27, 28,30,31,37,39,40	"	"	"	"
K14	J6	"	14	2,3,4,6,7,9,12,15, 16,17,20,22,24,25,27, 28,30,31,34,37,39,40	"	"	"	"
K15	J7	"	20	2,3,4,6,7,9,12,14,15, 16,17,22,24,25,27, 28,30,31,34,37,39,40	"	"	"	"
K16	J8	"	37	2,3,4,6,7,9,12,14,15, 16,17,20,22,24,25,27, 28,30,31,34,39,40	"	"	"	"
K17	$\overline{K1}$	"	25	2,3,4,6,7,9,12,14,15, 16,17,20,22,24,27, 28,30,31,34,37,39,40	"	"	"	"
K18	$\overline{K2}$	"	3	2,4,6,7,9,12,14,15, 16,17,20,22,24,25,27, 28,30,31,34,37,39,40	"	"	"	"
K19	$\overline{K3}$	"	28	2,3,4,6,7,9,12,14,15, 16,17,20,22,24,25,27, 30,31,34,37,39,40	"	"	"	"
K20	$\overline{K4}$	"	6	2,3,4,7,9,12,14,15, 16,17,20,22,24,25,27, 28,30,31,34,37,39,40	"	"	"	"

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K21	$\overline{K5}$	Input current, high level	15	2,3,4,6,7,9,12,14, 16,17,20,22,24,25,27, 28,30,31,34,37,39,40	0	50	µA	15
K22	$\overline{K6}$	"	12	2,3,4,6,7,9,14,15, 16,17,20,22,24,25,27, 28,30,31,34,37,39,40	"	"	"	"
K23	$\overline{K7}$	"	40	2,3,4,6,7,9,12,14,15, 16,17,20,22,24,25,27, 28,30,31,34,37,39	"	"	"	"
K24	$\overline{K8}$	"	17	2,3,4,6,7,9,12,14,15, 16,20,22,24,25,27, 28,30,31,34,37,39,40	"	"	"	"
K25	CP1-CP2	"	2	3,4,6,7,9,12,14,15, 16,17,20,22,24,25,27, 28,31,34,37,39,40	"	200	"	"
K26	CP7-CP8	"	39	2,3,4,6,7,9,12,14,15, 16,17,19,20,22,24,25,27, 28,31,34,37,40	"	200	"	"
K27	CP3	"	9	2,3,4,6,7,12,14,15, 16,17,20,22,24,25,27, 28,31,34,37,39,40	"	100	"	"
K28	CP4	"	22	2,3,4,6,7,9,12,14,15, 16,17,19,20,24,25,27, 28,31,34,37,39,40	"	"	"	"
K29	CP5	"	16	2,3,4,6,7,9,12,14,15, 17,19,20,22,24,25,27, 28,31,34,37,39,40	"	"	"	"
K30	CP6	"	31	2,3,4,6,7,9,12,14,15, 16,17,19,20,22,24,25,27, 28,34,37,39,40	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

FIGURE 15. Input current (high-level) test circuit for table XI.

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TABLE XII. Input current (low-level) test.

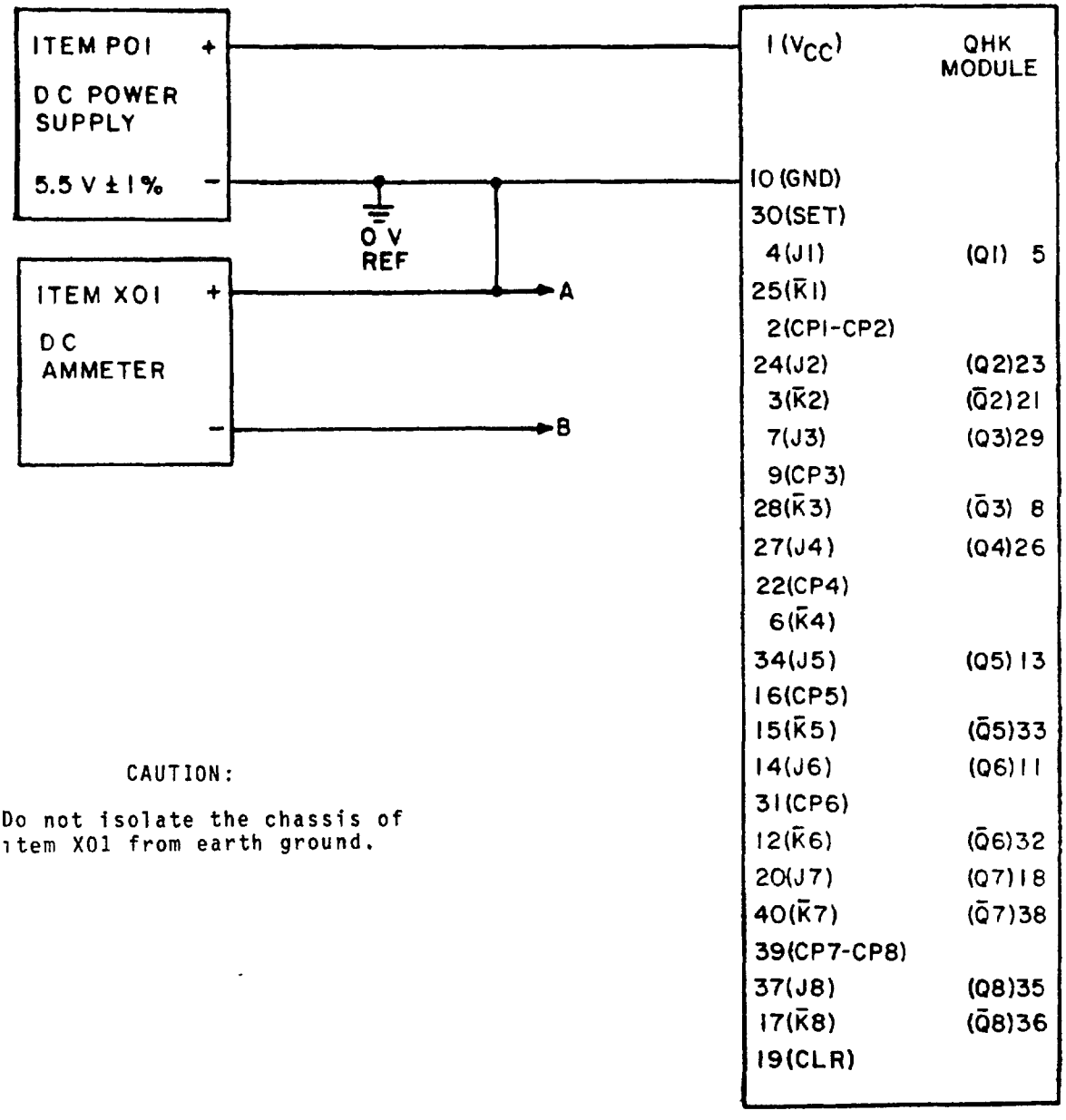
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	CLR Input current, low level	Open	19	-0.1	-8.0	mA	16
L2	CP1-CP2 "	3,4,19,24,25	2	"	"	"	"
L3	CP7-CP8 "	17,19,20,37,40	39	"	"	"	"
L4	SET "	Open	30	"	-4.0	"	"
L5	CP3 "	7,19,28	9	-0.10	-4.32	"	"
L6	CP4 "	6,19,27	22	"	"	"	"
L7	CP5 "	15,19,34	16	"	"	"	"
L8	CP6 "	12,14,19	31	"	"	"	"
L9	J1 "	19,30	4	"	-2.16	"	"
L10	J2 "	"	24	"	"	"	"
L11	J3 "	"	7	"	"	"	"
L12	J4 "	"	27	"	"	"	"
L13	J5 "	"	34	"	"	"	"
L14	J6 "	"	14	"	"	"	"
L15	J7 "	"	20	"	"	"	"
L16	J8 "	"	37	"	"	"	"
L17	$\overline{K}1$ "	19	25	"	"	"	"
L18	$\overline{K}2$ "	"	3	"	"	"	"
L19	$\overline{K}3$ "	"	28	"	"	"	"
L20	$\overline{K}4$ "	"	6	"	"	"	"

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TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L21	$\overline{K5}$	Input current, low level	19	15	-0.10	-2.16	mA	16
L22	$\overline{K6}$	"	"	12	"	"	"	"
L23	$\overline{K7}$	"	"	40	"	"	"	"
L24	$\overline{K8}$	"	"	17	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

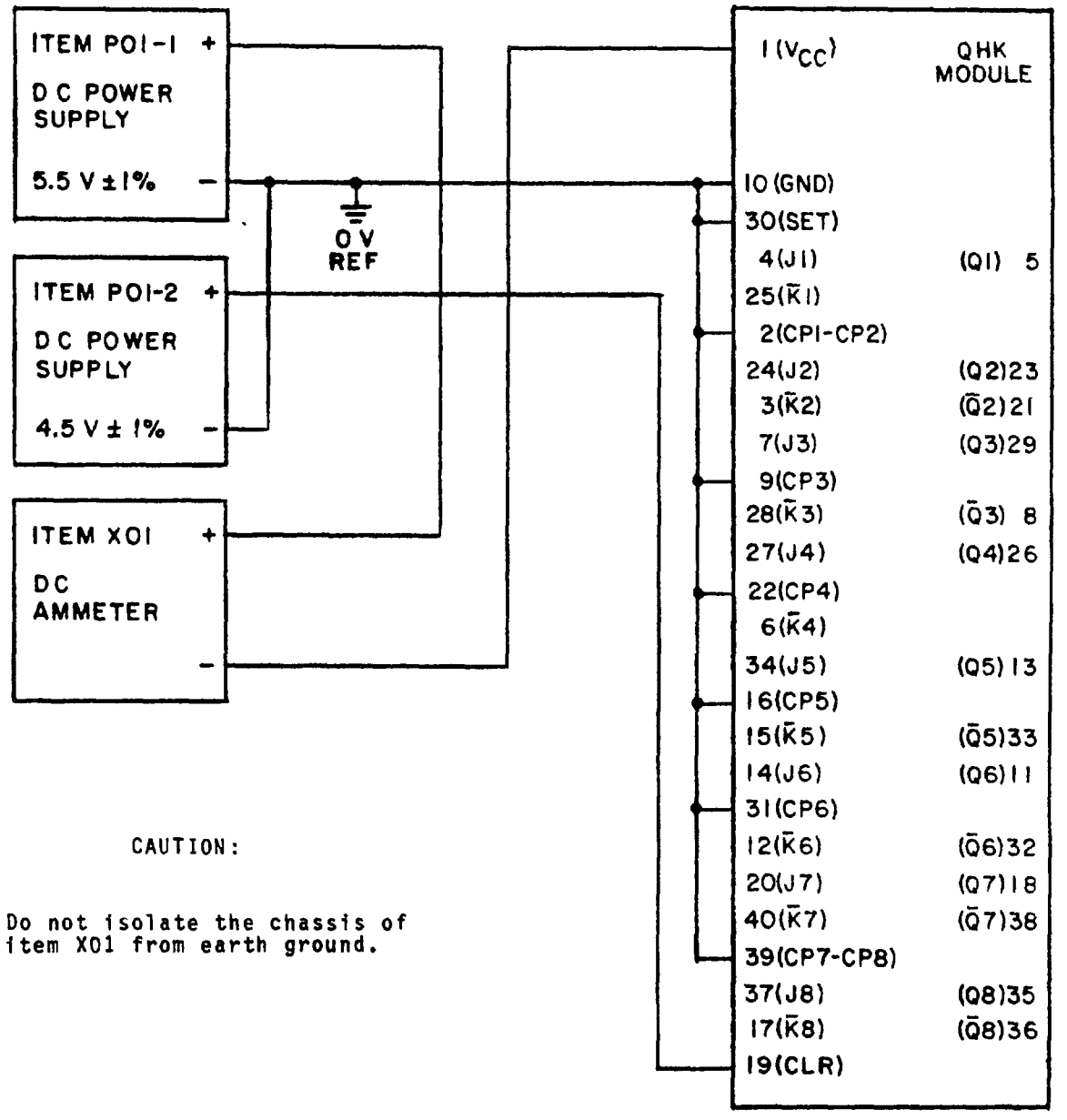
FIGURE 16. Input current (low-level) test circuit for table XII.

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TABLE XIII. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply current	0.1	262.0	mA	17

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 17. Power supply current test circuit for table XIII.

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) from clock test.

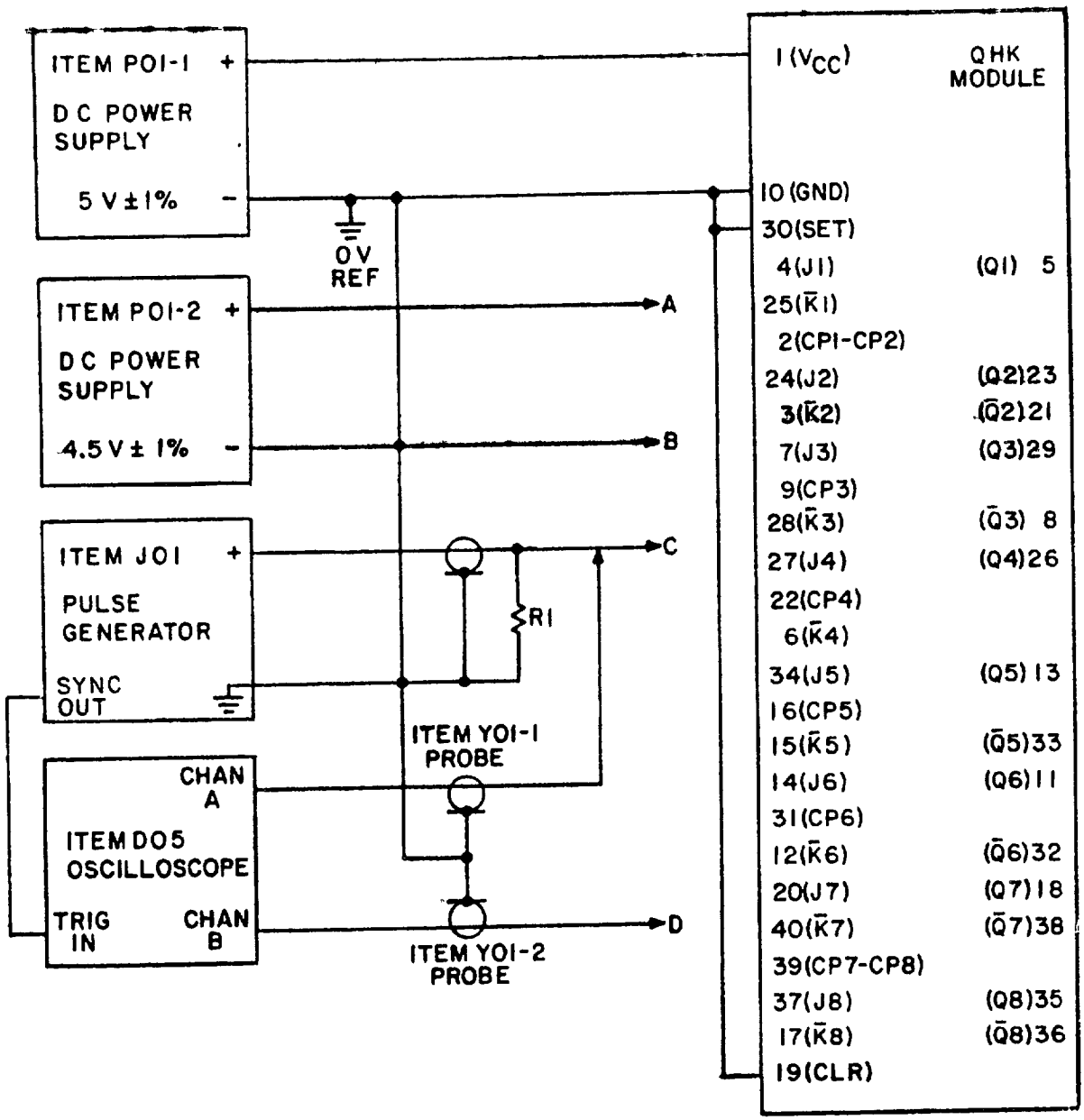
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Q1	Propagation delay time from clock (t_{PLH}), C_L = 50 pF	4	25	2	5	3	13	ns	18,19
N2	Q2	"	24	3	2	23	"	"	"	"
N3	Q3	"	7	28	9	29	"	"	"	"
N4	Q4	"	27	6	22	26	"	"	"	"
N5	Q5	"	34	15	16	13	"	"	"	"
N6	Q6	"	14	12	31	11	"	"	"	"
N7	Q7	"	20	40	39	18	"	"	"	"
N8	Q8	"	37	17	39	35	"	"	"	"
N9	$\overline{Q}2$	"	24	3	2	21	"	"	"	"
N10	$\overline{Q}3$	"	7	28	9	8	"	"	"	"
N11	$\overline{Q}5$	"	34	15	16	33	"	"	"	"
N12	$\overline{Q}6$	"	14	12	31	32	"	"	"	"
N13	$\overline{Q}7$	"	20	40	39	38	"	"	"	"
N14	$\overline{Q}8$	"	37	17	39	36	"	"	"	"
P1	Q1	Propagation delay time from clock (t_{PHL}), C_L = 50 pF	4	25	2	5	4	15	"	"
P2	Q2	"	24	3	2	23	"	"	"	"
P3	Q3	"	7	28	9	29	"	"	"	"
P4	Q4	"	27	6	22	26	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) from clock test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P5	Q5 Propagation delay time from clock (t_{PHL}), C_L = 50 pF	34	15	16	13	4	15	ns	18,19
P6	Q6 "	14	12	31	11	"	"	"	"
P7	Q7 "	20	40	39	18	"	"	"	"
P8	Q8 "	37	17	39	35	"	"	"	"
P9	$\overline{Q2}$ "	24	3	2	21	"	"	"	"
P10	$\overline{Q3}$ "	7	28	9	8	"	"	"	"
P11	$\overline{Q5}$ "	34	15	16	33	"	"	"	"
P12	$\overline{Q6}$ "	14	12	31	32	"	"	"	"
P13	$\overline{Q7}$ "	20	40	39	38	"	"	"	"
P14	$\overline{Q8}$ "	37	17	39	36	"	"	"	"

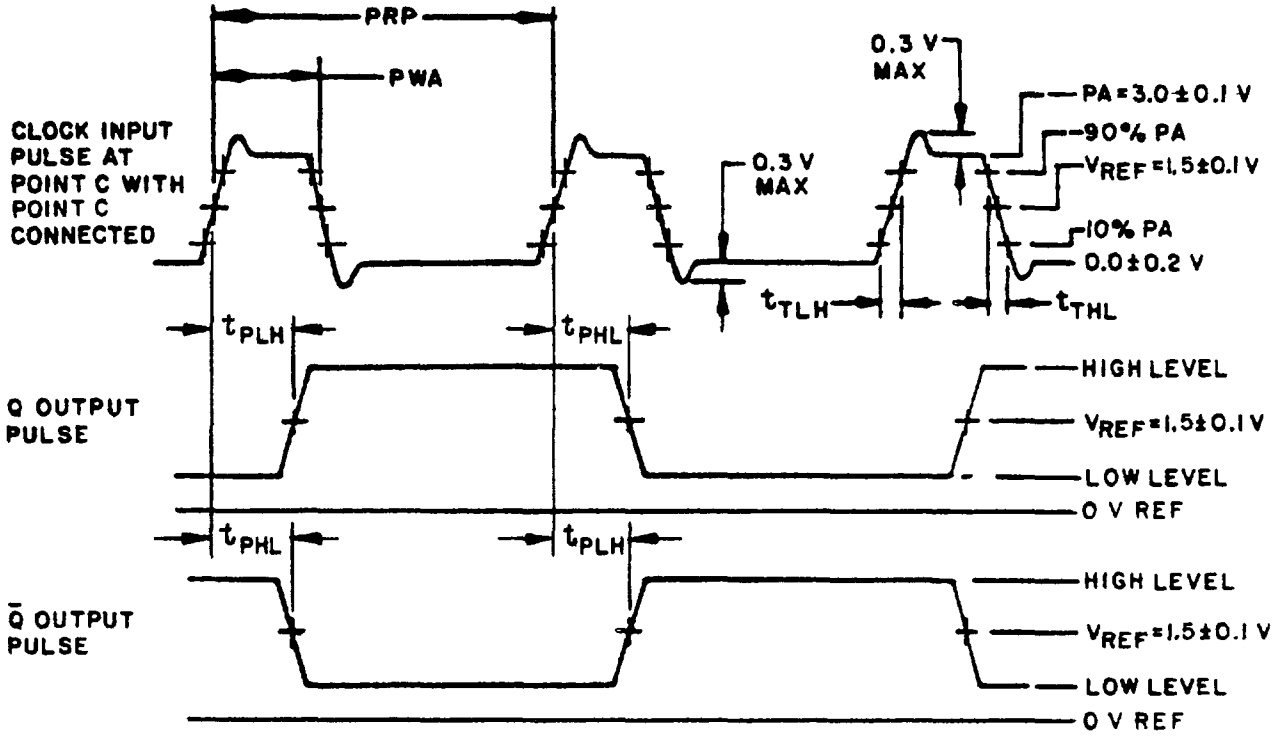
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:48Z
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load outputs as shown on figure 14.

FIGURE 18. Propagation delay time (t_{PLH} and t_{PHL}) from clock test circuit for table XIV.

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NOTES:
1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns.
   PWA = 250 ±50 ns.
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
2. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.5 V before making measurements.
3. All voltages are referenced to ground.

FIGURE 19. Propagation delay time (t_{PLH} and t_{PHL}) from clock test circuit waveforms for table XIV.

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TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) from set and clear and minimum set and clear pulse width check test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q1	Q1	Propagation delay time from set (t_{PLH}), C_L = 50 pF	30	19	5	4	14	ns	20,21
Q2	Q2	"	"	"	23	"	"	"	"
Q3	Q3	"	"	"	29	"	"	"	"
Q4	Q4	"	"	"	26	"	"	"	"
Q5	Q5	"	"	"	13	"	"	"	"
Q6	Q6	"	"	"	11	"	"	"	"
Q7	Q7	"	"	"	18	"	"	"	"
Q8	Q8	"	"	"	35	"	"	"	"
Q9	Q1	"	"	"	5	"	"	"	"
Q10	Q2	"	"	"	23	"	"	"	"
Q11	Q3	"	"	"	29	"	"	"	"
Q12	Q4	"	"	"	26	"	"	"	"
Q13	Q5	"	"	"	13	"	"	"	"
Q14	Q6	"	"	"	11	"	"	"	"
Q15	Q7	"	"	"	18	"	"	"	"
Q16	Q8	"	"	"	35	"	"	"	"
R1	$\overline{Q2}$	Propagation delay time from set (t_{PHL}), C_L = 50 pF	"	"	21	6	21	"	"
R2	$\overline{Q3}$	"	"	"	8	"	"	"	"
R3	$\overline{Q5}$	"	"	"	33	"	"	"	"
R4	$\overline{Q6}$	"	"	"	32	"	"	"	"
R5	$\overline{Q7}$	"	"	"	38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:48Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) from set and clear and minimum set and clear pulse width check test - Continued.

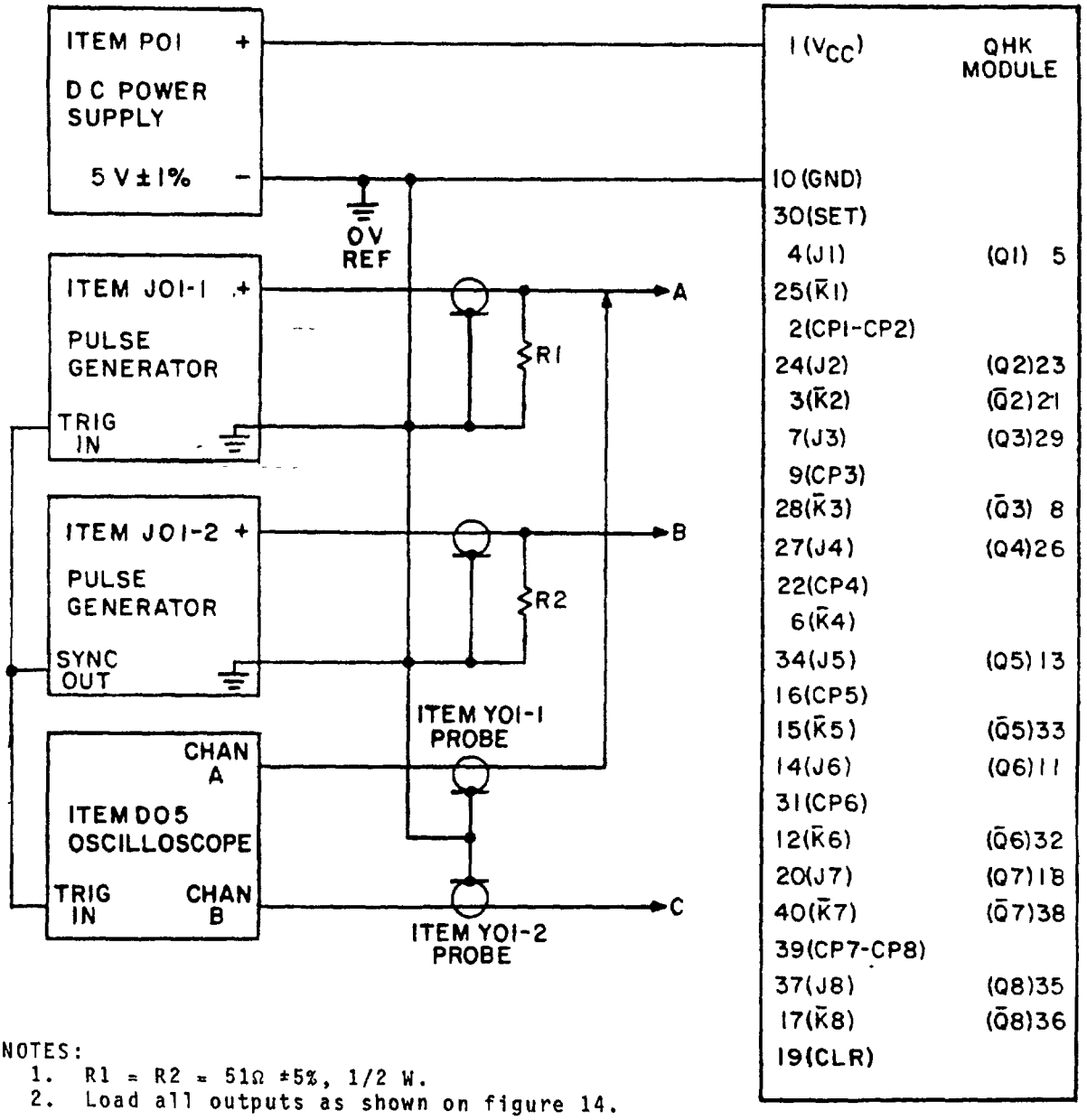
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R6	$\overline{Q}8$	Propagation delay time from set (t_{PHL}), C_L = 50 pF	30	19	36	6	21	ns	20,21
R7	$\overline{Q}2$	"	"	"	21	"	"	"	"
R8	$\overline{Q}3$	"	"	"	8	"	"	"	"
R9	$\overline{Q}5$	"	"	"	33	"	"	"	"
R10	$\overline{Q}6$	"	"	"	32	"	"	"	"
R11	$\overline{Q}7$	"	"	"	38	"	"	"	"
R12	$\overline{Q}8$	"	"	"	36	"	"	"	"
S1	$\overline{Q}2$	Propagation delay time from clear (t_{PLH}), C_L = 50 pF	19	30	21	4	14	"	20,22
S2	$\overline{Q}3$	"	"	"	8	"	"	"	"
S3	$\overline{Q}5$	"	"	"	33	"	"	"	"
S4	$\overline{Q}6$	"	"	"	32	"	"	"	"
S5	$\overline{Q}7$	"	"	"	38	"	"	"	"
S6	$\overline{Q}8$	"	"	"	36	"	"	"	"
S7	$\overline{Q}2$	"	"	"	21	"	"	"	"
S8	$\overline{Q}3$	"	"	"	8	"	"	"	"
S9	$\overline{Q}5$	"	"	"	33	"	"	"	"
S10	$\overline{Q}6$	"	"	"	32	"	"	"	"
S11	$\overline{Q}7$	"	"	"	38	"	"	"	"
S12	$\overline{Q}8$	"	"	"	36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:48Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) from set and clear and minimum set and clear pulse width check test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T1	Q1	Propagation delay time from clear (t_{PHL}), C_L = 50 pF	19	30	5	6	21	ns	20,22
T2	Q2	"	"	"	23	"	"	"	"
T3	Q3	"	"	"	29	"	"	"	"
T4	Q4	"	"	"	26	"	"	"	"
T5	Q5	"	"	"	13	"	"	"	"
T6	Q6	"	"	"	11	"	"	"	"
T7	Q7	"	"	"	18	"	"	"	"
T8	Q8	"	"	"	35	"	"	"	"
T9	Q1	"	"	"	5	"	"	"	"
T10	Q2	"	"	"	23	"	"	"	"
T11	Q3	"	"	"	29	"	"	"	"
T12	Q4	"	"	"	26	"	"	"	"
T13	Q5	"	"	"	13	"	"	"	"
T14	Q6	"	"	"	11	"	"	"	"
T15	Q7	"	"	"	18	"	"	"	"
T16	Q8	"	"	"	35	"	"	"	"

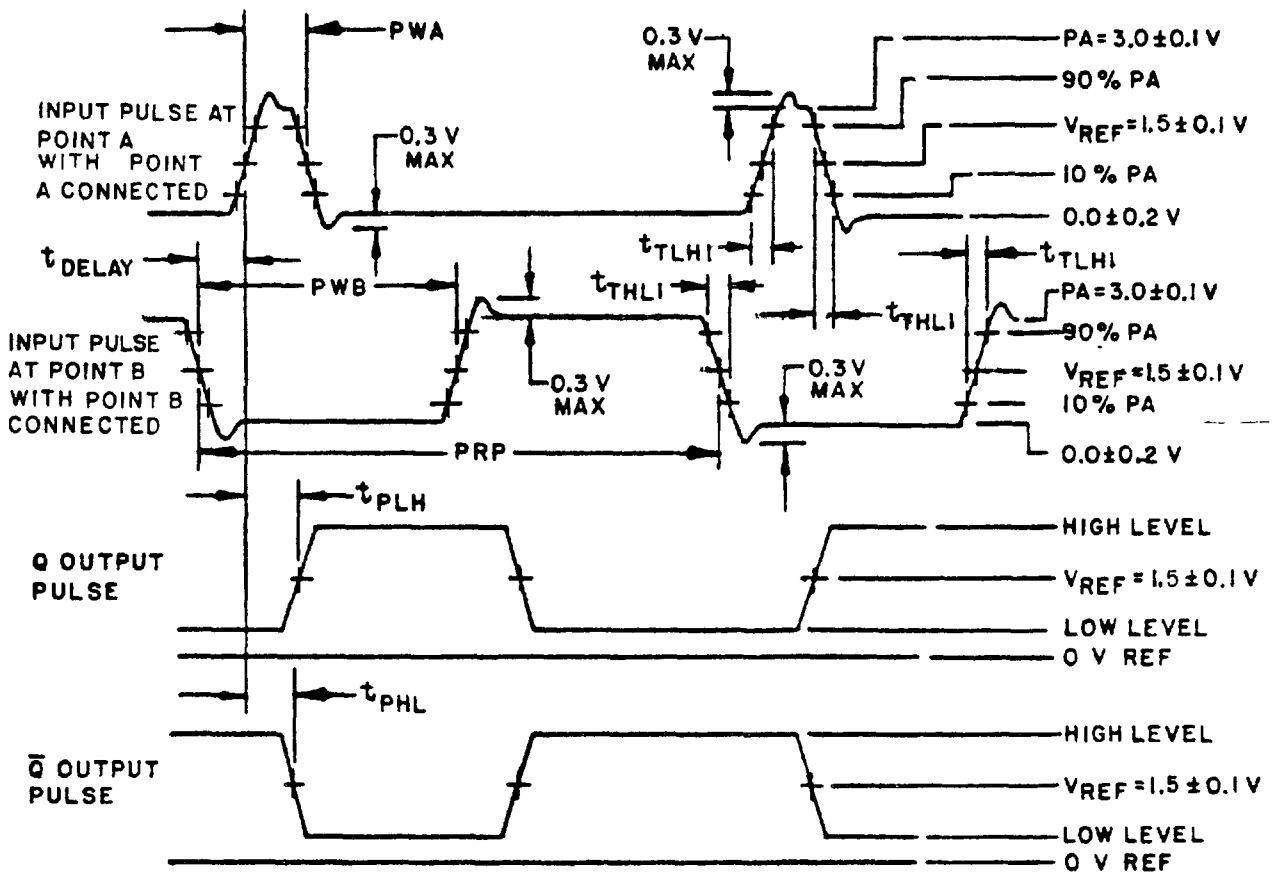
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs as shown on figure 14.

FIGURE 20. Propagation delay time (t_{PLH} and t_{PHL}) from set and clear and minimum set and clear pulse width check test circuit for table XV.

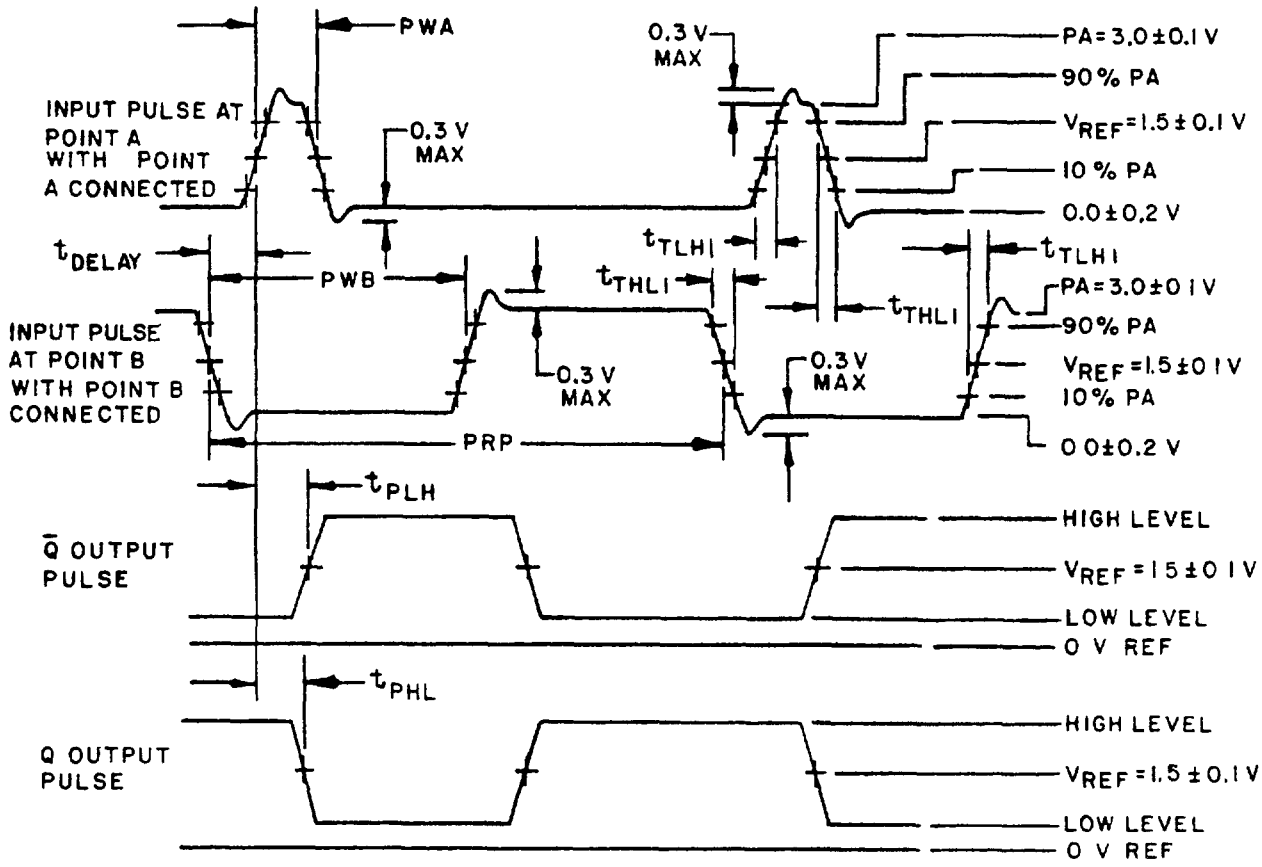
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:48Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters (item J01-1):
   PWA = See note 3.
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
2. Input pulse parameters (item J01-2):
   PRP = 1,000 ±100 ns.
   PWB = 250 ±25 ns.
   $t_{DELAY} = 50 \pm 5$ ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
3. Minimum set and clear pulse width check:
   a. For test conditions Q1-Q8 and R1-R6:
      (1) Adjust item J01-1 to obtain PWA = 16 +2, -0 ns (minimum set and clear pulse width).
      (2) The pulse width requirements must be verified before making measurements.
   b. For test conditions Q9-Q16 and R7-R12:
      (1) Adjust item J01-1 to obtain PWA = 6 +1, -0 ns (minimum set and clear pulse width).
      (2) The pulse width requirements must be verified before making measurements.
4. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.5 V before making measurements.
5. All voltages are referenced to ground.

FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) from set and minimum set pulse width check test circuit waveforms for table XV.

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NOTES:
1. Input pulse parameters (item J01-1):
   PWA = See note 3.
   $t_{TLH1} = t_{THL1} = 6 +1, -0$ ns.
2. Input pulse parameters (item J01-2):
   PRP = 1,000 ±100 ns.
   PWB = 250 ±25 ns.
   $t_{DELAY} = 50 \pm 5$ ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
3. Minimum set and clear pulse width check:
   a. For test conditions S1-S6 and T1-T8:
      (1) Adjust item J01-1 to obtain PWA = 16 +2, -0 ns (minimum set and clear pulse width).
      (2) The pulse width requirements must be verified before making measurements.
   b. For test conditions S7-S12 and T9-T16:
      (1) Adjust item J01-1 to obtain PWA = 6 +1, -0 ns (minimum set and clear pulse width).
      (2) The pulse width requirements must be verified before making measurements.
4. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.5 V before making measurements.
5. All voltages are referenced to ground.

FIGURE 22. Propagation delay tiem (t_{PLH} and t_{PHL}) from clear and minimum clear pulse width check test circuit waveforms for table XV.

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TABLE XVI. Operating frequency and minimum clock pulse width check test.

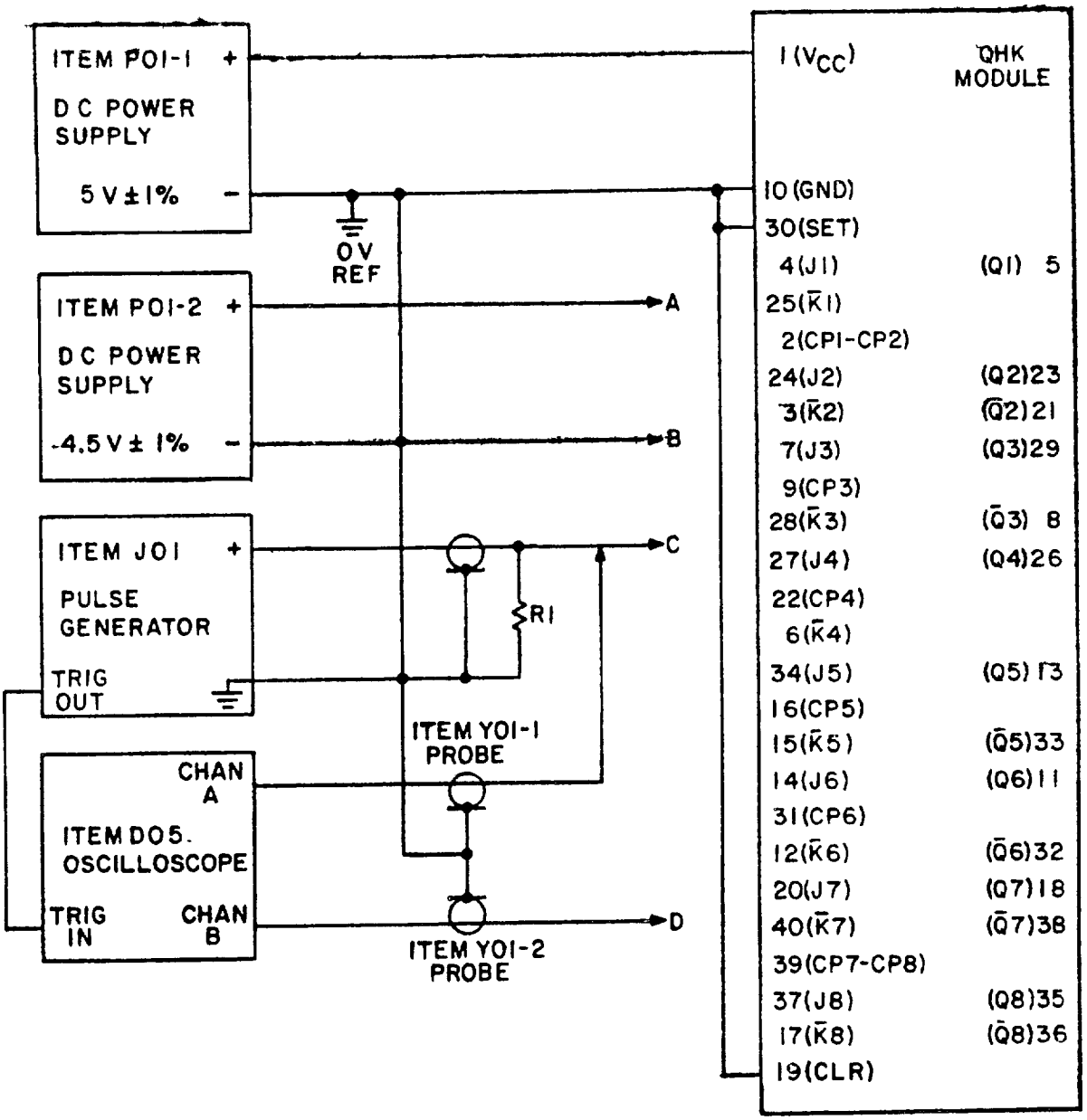
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	Q1 Operating frequency and minimum clock pulse width check, C_L = 50 pF	4	25	2	5	90	110	ns	23,24
U2	Q2 "	24	3	2	23	"	"	"	"
U3	Q3 "	7	28	9	29	"	"	"	"
U4	Q4 "	27	6	22	26	"	"	"	"
U5	Q5 "	34	15	16	13	"	"	"	"
U6	Q6 "	14	12	31	11	"	"	"	"
U7	Q7 "	20	40	39	18	"	"	"	"
U8	Q8 "	37	17	39	35	"	"	"	"
U9	$\overline{Q2}$ "	24	3	2	21	"	"	"	"
U10	$\overline{Q3}$ "	7	28	9	8	"	"	"	"
U11	$\overline{Q5}$ "	34	15	16	33	"	"	"	"
U12	$\overline{Q6}$ "	14	12	31	32	"	"	"	"
U13	$\overline{Q7}$ "	20	40	39	38	"	"	"	"
U14	$\overline{Q8}$ "	37	17	39	36	"	"	"	"
U15	Q1 "	4	25	2	5	20	30	"	23,25
U16	Q2 "	24	3	2	23	"	"	"	"
U17	Q3 "	7	28	9	29	"	"	"	"
U18	Q4 "	27	6	22	26	"	"	"	"
U19	Q5 "	34	15	16	13	"	"	"	"

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TABLE XVI. Operating frequency and minimum clock pulse width check test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U20	Q6 Operating frequency and minimum clock pulse width check, C_L = 50 pF	14	12	31	11	20	30	ns	23,25
U21	Q7 "	20	40	39	18	"	"	"	"
U22	Q8 "	37	17	39	35	"	"	"	"
U23	$\overline{Q}$2 "	24	3	2	21	"	"	"	"
U24	$\overline{Q}$3 "	7	28	9	8	"	"	"	"
U25	$\overline{Q}$5 "	34	15	16	33	"	"	"	"
U26	$\overline{Q}$6 "	14	12	31	32	"	"	"	"
U27	$\overline{Q}$7 "	20	40	39	38	"	"	"	"
U28	$\overline{Q}$8 "	37	17	39	36	"	"	"	"

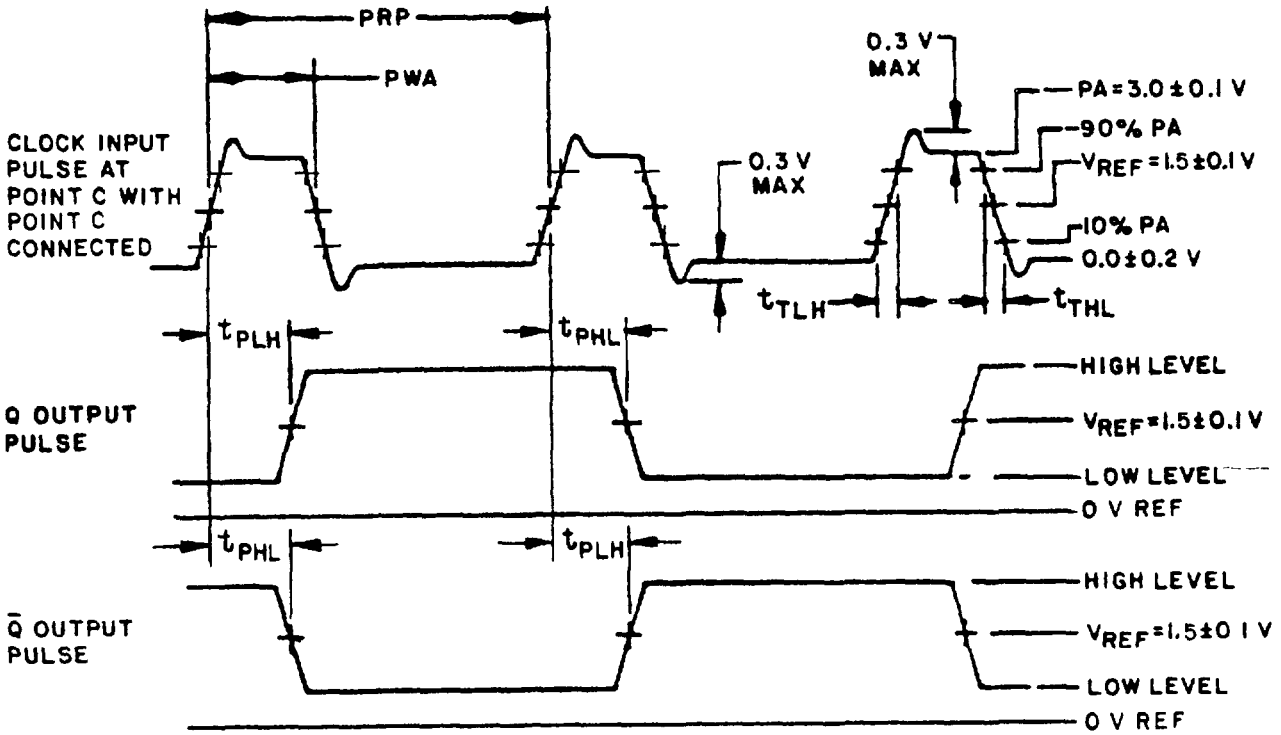
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs as shown on figure 14.

FIGURE 23. Operating frequency and minimum clock pulse width check test circuit for table XVI.

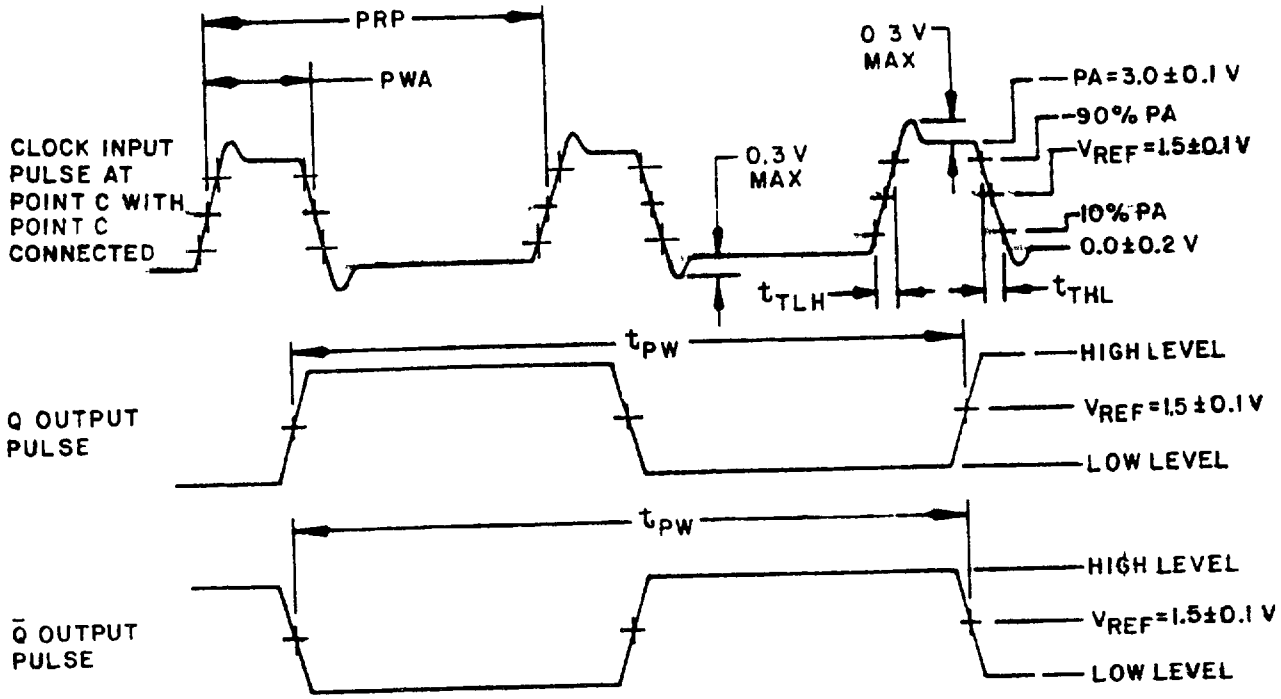
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:48Z
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NOTES:
1. Clock input pulse parameters:
   PRP = 50 ±2 ns.
   PWA = 25 ±1 ns.
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns.
2. Verify that all outputs have a high level $\geq$2.5 V and a low level $\leq$0.5 V before making measurements.
3. All voltages are referenced to ground.

FIGURE 24. Operating frequency and minimum clock pulse width check test circuit waveforms for table XVI.

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NOTES:
1. Clock input pulse parameters:
   PRP = 13 +0, -1 ns (75 +2, -0 MHz).
   PWA = 8 ±1 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
2. Verify that all outputs have a high level $\geq$2.5 V and a low level $\leq$0.5 V before making measurements.
3. All voltages are referenced to ground.

FIGURE 25. Operating frequency and minimum clock pulse width check test circuit waveforms for table XVI.

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TABLE XVII. Output crosstalk (low-level and high-level) test.

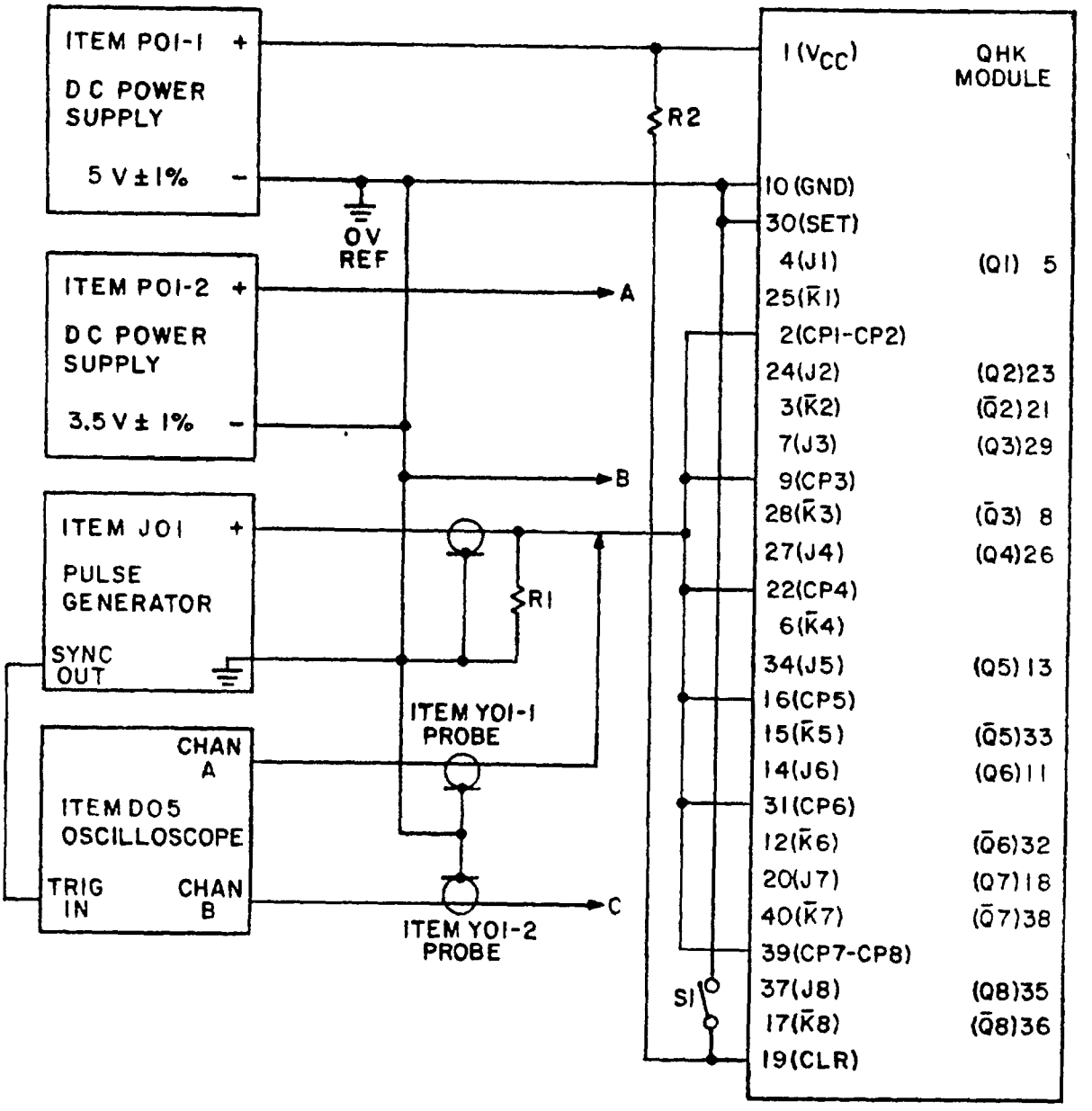
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V1	Q1 Output crosstalk, low level	7,14,20,24,27, 34,37	3,4,6,12,15, 17,25,28,40	5	0	400	mV	26,27
V2	Q2 "	4,7,14,20,27, 34,37	3,6,12,15,17, 24,25,28,40	23	"	"	"	"
V3	Q3 "	4,14,20,24,27, 34,37	3,6,7,12,15, 17,25,28,40	29	"	"	"	"
V4	Q4 "	4,7,14,20,24, 34,37	3,6,12,15,17, 25,27,28,40	26	"	"	"	"
V5	Q5 "	4,7,14,20,24, 27,37	3,6,12,15,17, 25,28,34,40	13	"	"	"	"
V6	Q6 "	4,7,20,24,27, 34,37	3,6,12,14,15, 17,25,28,40	11	"	"	"	"
V7	Q7 "	4,7,14,24,27, 34,37	3,6,12,15,17, 20,25,28,40	18	"	"	"	"
V8	Q8 "	4,7,14,20,24, 27,34	3,6,12,15,17, 25,28,37,40	35	"	"	"	"
V9	$\overline{Q2}$ "	3,4,7,14,20,24, 27,34,37	6,12,15,17, 25,28,40	21	"	"	"	"
V10	$\overline{Q3}$ "	4,7,14,20,24, 27,28,34,37	3,6,12,15,17, 25,40	8	"	"	"	"
V11	$\overline{Q5}$ "	4,7,14,15,20, 24,27,34,37	3,6,12,17,25, 28,40	33	"	"	"	"
V11	$\overline{Q5}$ "	4,7,14,15,20, 24,27,34,37	3,6,12,17,25, 28,40	33	"	"	"	"
V12	$\overline{Q6}$ "	4,7,12,14,20, 24,27,34,37	3,6,15,17,25, 28,40	32	"	"	"	"
V13	$\overline{Q7}$ "	4,7,14,20,24, 27,34,37,40	3,6,12,15,17, 25,28	38	"	"	"	"
V14	$\overline{Q8}$ "	4,7,14,17,20, 24,27,34,37	3,6,12,15,25, 28,40	36	"	"	"	"

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W1	Q1 Output crosstalk, high level	4,7,14,20,24, 25,27,34,37	3,6,12,15,17, 28,40	5	0	400	mV	26,27
W2	Q2 "	3,4,7,14,20,24, 27,34,37	6,12,15,17, 25,28,40	23	"	"	"	"
W3	Q3 "	4,7,14,20,24, 27,28,34,37	3,6,12,15,17, 25,40	29	"	"	"	"
W4	Q4 "	4,6,7,14,20,24, 27,34,37	3,12,15,17, 25,28,40	26	"	"	"	"
W5	Q5 "	4,7,14,15,20, 24,27,34,37	3,6,12,17,25, 28,40	13	"	"	"	"
W6	Q6 "	4,7,12,14,20, 24,27,34,37	3,6,15,17,25, 28,40	11	"	"	"	"
W7	Q7 "	4,7,14,20,24, 27,34,37,40	3,6,12,15,17, 25,28	18	"	"	"	"
W8	Q8 "	4,7,14,17,20, 24,27,34,37	3,6,12,15,25, 28,40	35	"	"	"	"
W9	$\overline{Q2}$ "	4,7,14,20,27, 34,37	3,6,12,15,17, 24,25,28,40	21	"	"	"	"
W10	$\overline{Q3}$ "	4,14,20,24,27, 34,37	3,6,7,12,15, 17,25,28,40	8	"	"	"	"
W11	$\overline{Q5}$ "	4,7,14,20,24, 27,37	3,6,12,15,17, 25,28,34,40	33	"	"	"	"
W12	$\overline{Q6}$ "	4,7,20,24,27, 34,37	3,6,12,14,15, 17,25,28,40	32	"	"	"	"
W13	$\overline{Q7}$ "	4,7,14,24,27, 34,37	3,6,12,15,17, 20,25,28,40	38	"	"	"	"
W14	$\overline{Q8}$ "	4,7,14,20,24, 27,34	3,6,12,15,17, 25,28,37,40	36	"	"	"	"

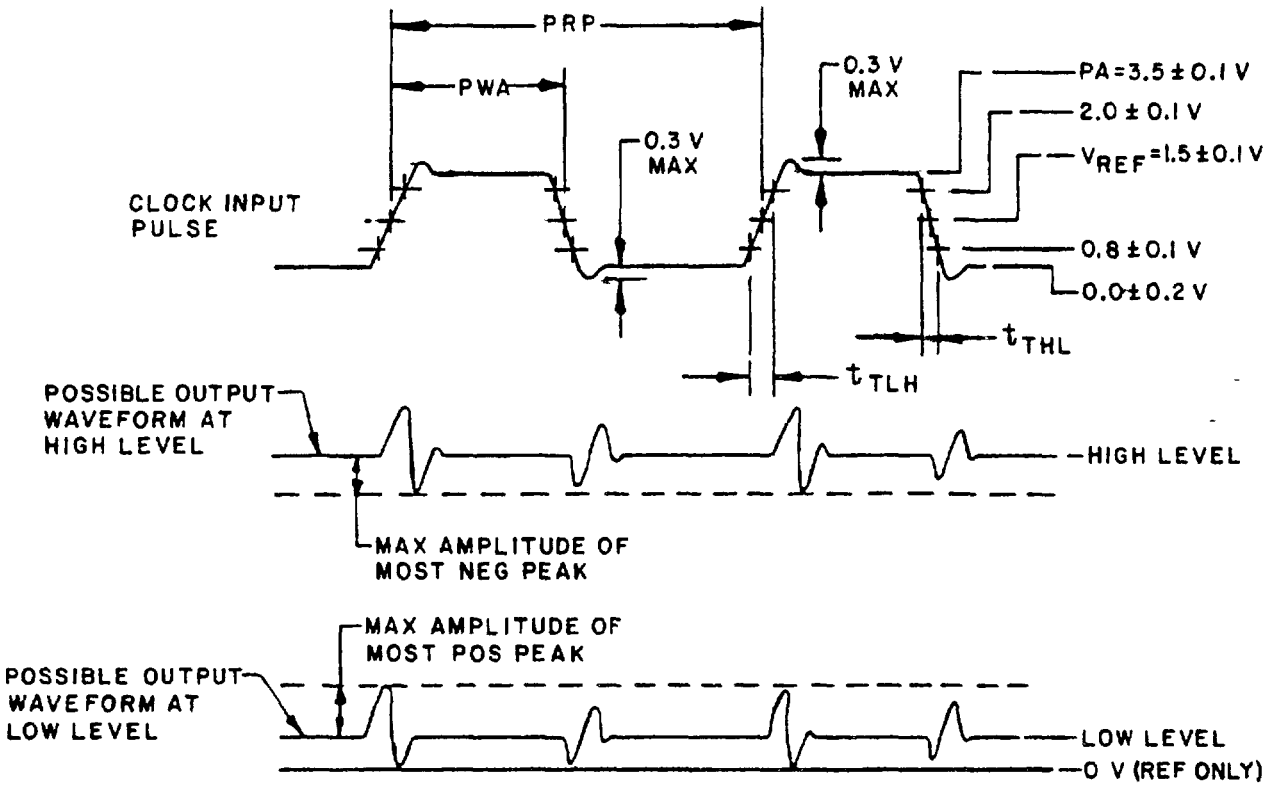
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 1 kΩ ±5%, 1/4 W.
3. S1 = SPST switch.
4. Load all outputs with load as specified on figure 14.
5. After making all circuit connections for a test conditions open switch S1 for a minimum of 100 ns then close switch S1.

FIGURE 26. Output crosstalk (low-level and high-level) test circuit for table XVII.

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NOTES:
1. Clock input pulse parameters:
   a. PRP = 1,000 ±200 ns.
   b. PWA = 500 ±100 ns.
   c. $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (measured at the 0.8 V and 2.0 V points of the input pulse).
   d. For each test condition with the module connected to item J01, adjust item J01 for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making each measurement.
2. The lead length of electrical conductors between resistor R1 and input pins, between items Y01-1 and Y01-2 and input and output pins, between load circuits and output pins, and between ground leads of items Y01-1 and Y01-2 and test fixtures and module ground pins shall be less than 2 inches (50.8 mm).

FIGURE 27. Output crosstalk (low-level and high-level) test circuit waveforms for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
X	Power supply transient amplitude	0	400	mV	28,29

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y	Power supply filter capacitance	Open	1	1.6	3.0	µF	30

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TABLE XX. Input capacitance test.

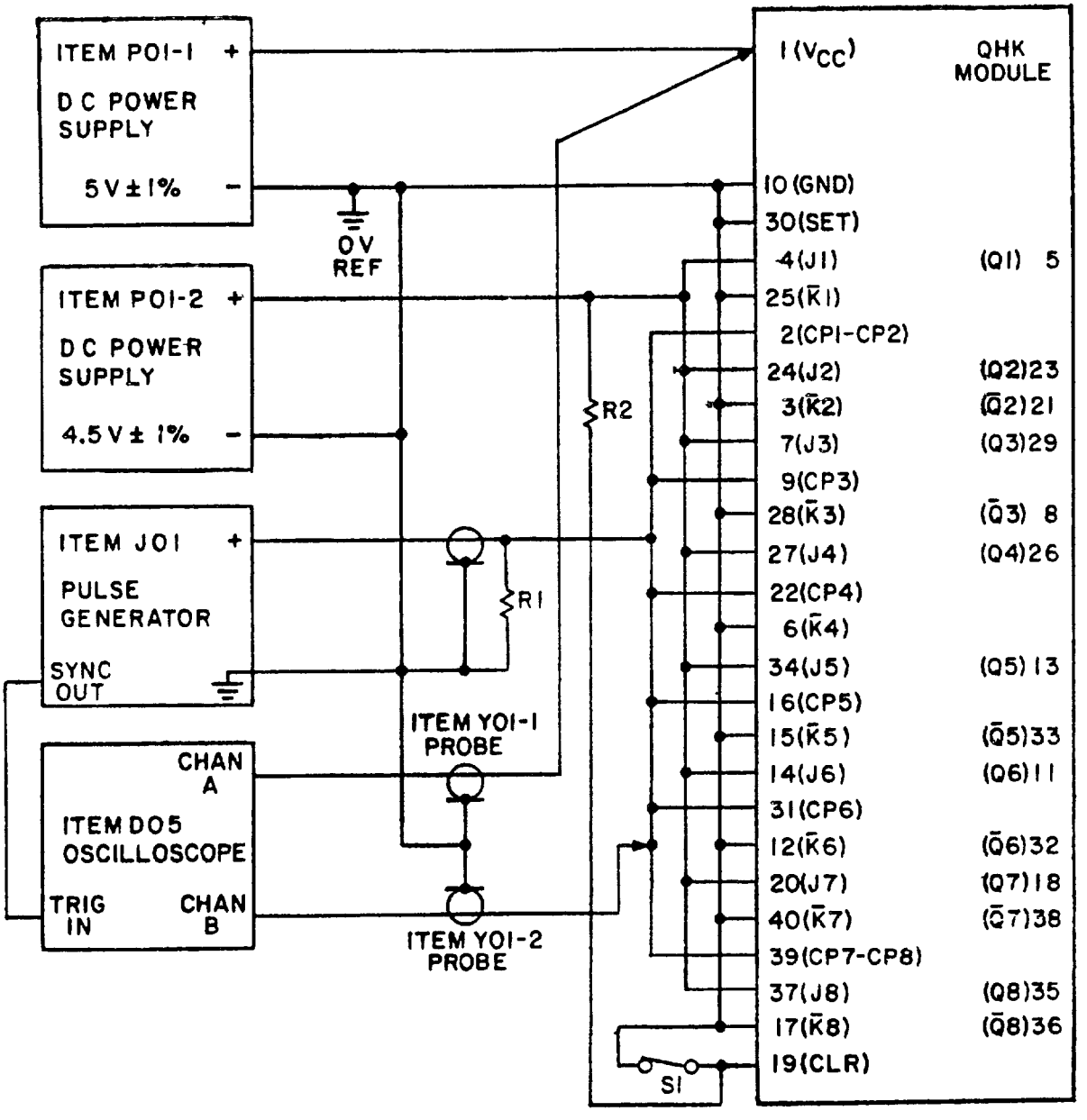
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z1	CLR Input capacitance	19	Open	5	60	pF	30
Z2	SET "	30	"	"	30	"	"
Z3	CP1-CP2 "	2	"	"	"	"	"
Z4	CP7-CP8 "	39	"	"	"	"	"
Z5	CP3 "	9	"	"	15	"	"
Z6	CP4 "	22	"	"	"	"	"
Z7	CP5 "	16	"	"	"	"	"
Z8	CP6 "	31	"	"	"	"	"
Z9	J1 "	4	"	"	"	"	"
Z10	J2 "	24	"	"	"	"	"
Z11	J3 "	7	"	"	"	"	"
Z12	J4 "	27	"	"	"	"	"
Z13	J5 "	34	"	"	"	"	"
Z14	J6 "	14	"	"	"	"	"
Z15	J7 "	20	"	"	"	"	"
Z16	J8 "	37	"	"	"	"	"
Z17	$\overline{K1}$ "	25	"	"	"	"	"
Z18	$\overline{K2}$ "	3	"	"	"	"	"
Z19	$\overline{K3}$ "	28	"	"	"	"	"
Z20	$\overline{K4}$ "	6	"	"	"	"	"

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Z21	$\overline{K}5$	Input capacitance	15	Open	5	15	pF	30
Z22	$\overline{K}6$	"	12	"	"	"	"	"
Z23	$\overline{K}7$	"	40	"	"	"	"	"
Z24	$\overline{K}8$	"	17	"	"	"	"	"

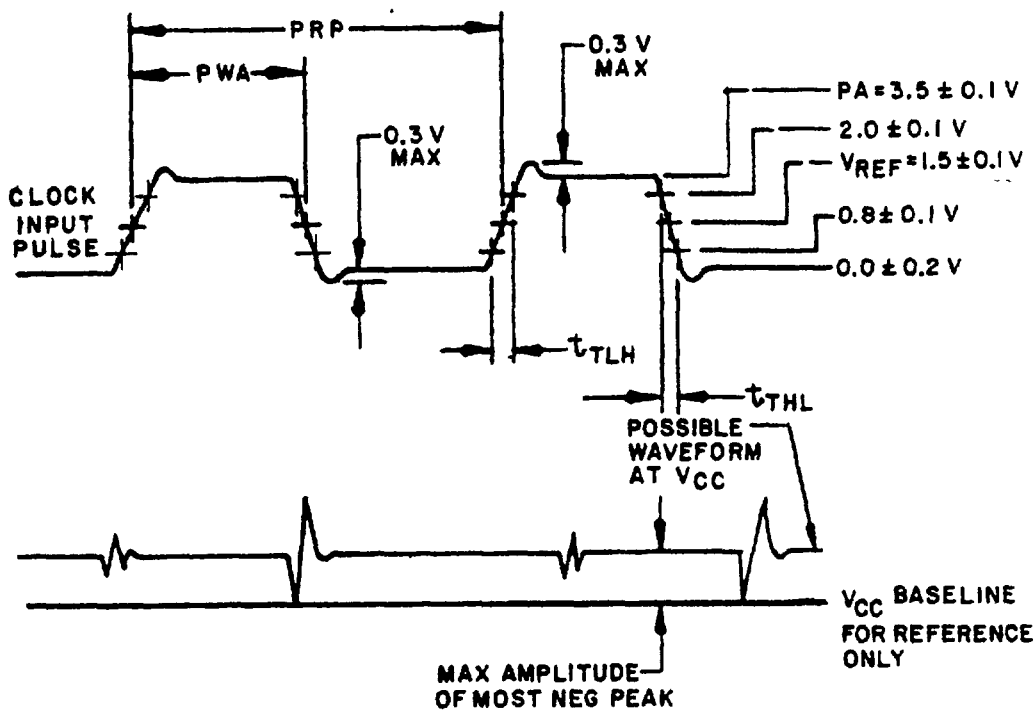
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 1 kΩ ±5%, 1/4 W.
3. S1 = SPST switch.
5. After making all circuit connections for a test condition open switch S1 for a minimum of 100 ns then close switch S1 and make measurement.

FIGURE 28. Power supply transient amplitude test circuit for table XVIII.

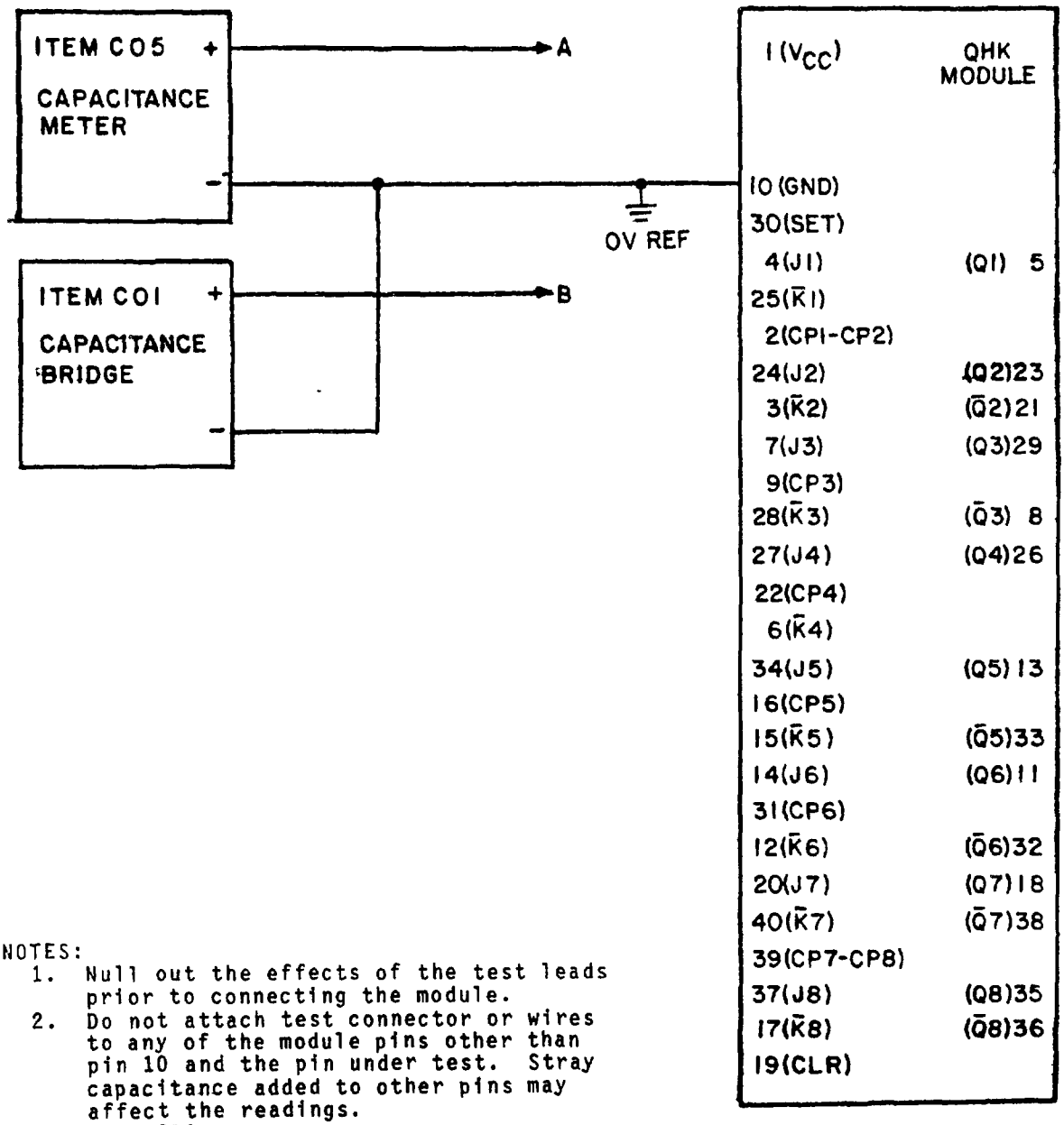
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NOTES:
1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns.
   PWA = 500 ±100 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.
2. For each test condition with the module connected to item J01, adjust item J01 for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making each measurement.

FIGURE 29. Power supply transient amplitude test circuit waveforms for table XVIII.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 30. Power supply filter and input capacitance test circuit for tables XIX and XX.

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TABLE XXI. Setup time test.

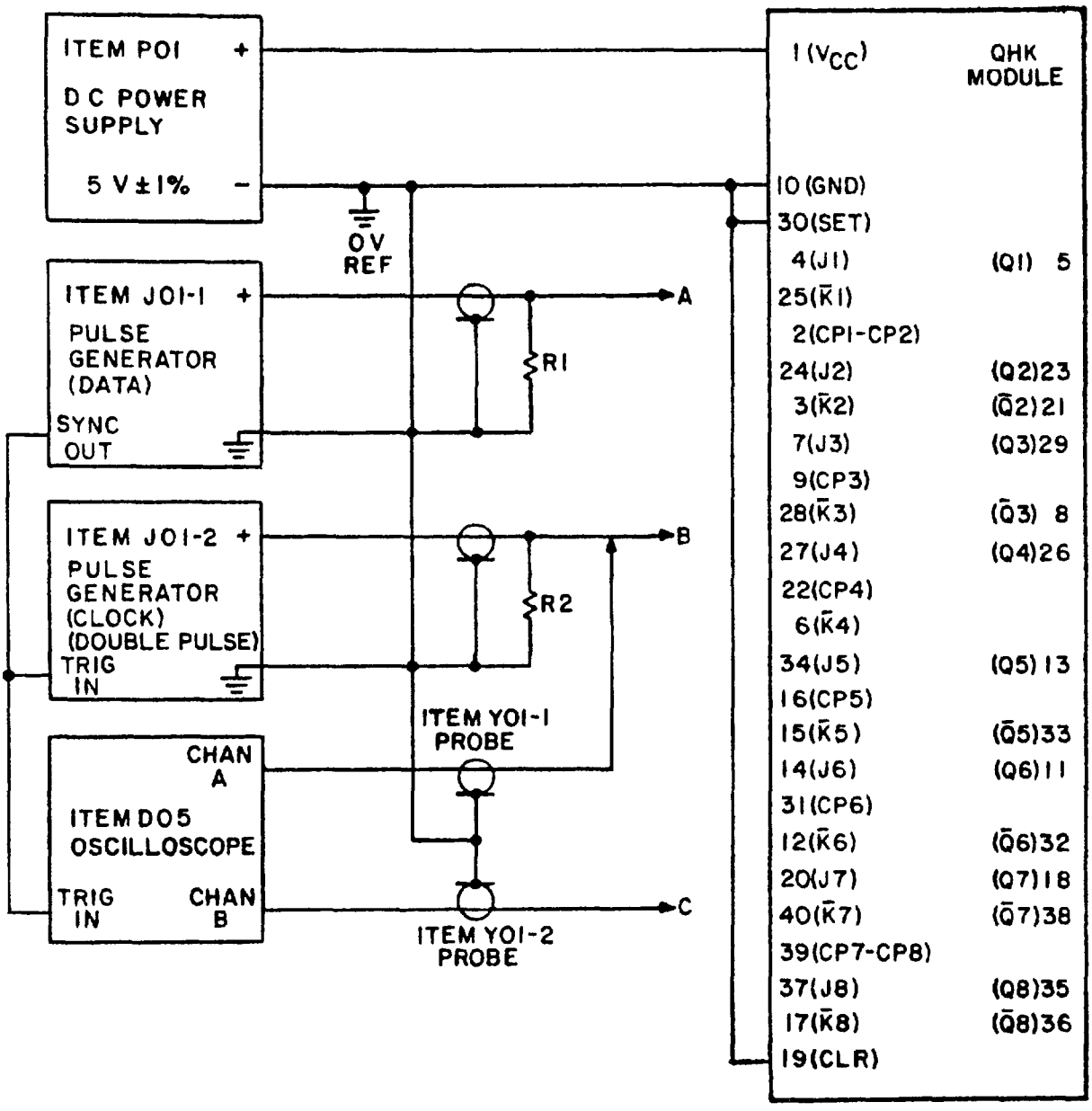
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
AA1	Q1 Setup time, C_L = 50 pF	4,25	2	5	80	120	ns	31,32
AA2	Q2 "	3,24	2	23	"	"	"	"
AA3	Q3 "	7,28	9	29	"	"	"	"
AA4	Q4 "	6,27	22	26	"	"	"	"
AA5	Q5 "	15,34	16	13	"	"	"	"
AA6	Q6 "	12,14	31	11	"	"	"	"
AA7	Q7 "	20,40	39	18	"	"	"	"
AA8	Q8 "	17,37	39	35	"	"	"	"
AA9	$\overline{Q2}$ "	3,24	2	21	"	"	"	"
AA10	$\overline{Q3}$ "	7,28	9	8	"	"	"	"
AA11	$\overline{Q5}$ "	15,34	16	33	"	"	"	"
AA12	$\overline{Q6}$ "	12,14	31	32	"	"	"	"
AA13	$\overline{Q7}$ "	20,40	39	38	"	"	"	"
AA14	$\overline{Q8}$ "	17,37	39	36	"	"	"	"
AA15	Q1 "	4,25	2	5	"	"	"	"
AA16	Q2 "	3,24	2	23	"	"	"	"
AA17	Q3 "	7,28	9	29	"	"	"	"
AA18	Q4 "	6,27	22	26	"	"	"	"
AA19	Q5 "	15,34	16	13	"	"	"	"
AA20	Q6 "	12,14	31	11	"	"	"	"

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TABLE XXI. Setup time test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
AA21	Q7 Setup time, C_L = 50 pF	20,40	39	18	80	120	ns	31,32
AA22	Q8 "	17,37	39	35	"	"	"	"
AA23	$\overline{Q2}$ "	3,24	2	21	"	"	"	"
AA24	$\overline{Q3}$ "	7,28	9	8	"	"	"	"
AA25	$\overline{Q5}$ "	15,34	16	33	"	"	"	"
AA26	$\overline{Q6}$ "	12,14	31	32	"	"	"	"
AA27	$\overline{Q7}$ "	20,40	39	38	"	"	"	"
AA28	$\overline{Q8}$ "	17,37	39	36	"	"	"	"

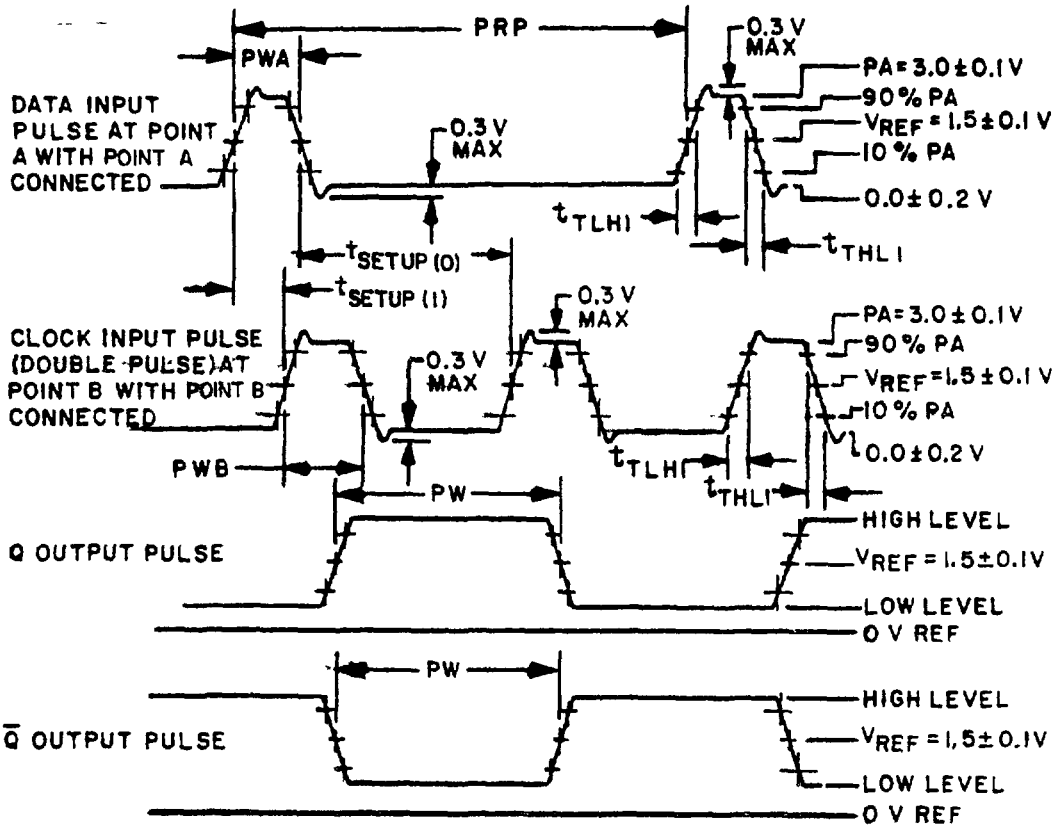
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Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs as shown on figure 14.

FIGURE 31. Setup time test circuit for table XXI.

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NOTES:
1. Data input pulse parameters:
   a. PRP = 200 ±20 ns.
   b. $t_{TLH1} = t_{THL1} = 3 \pm 1$ ns.
   c. PWA = 100 ±10 ns.
2. Clock input pulse parameters:
   a. Item J01-2 is operated in double pulse mode.
   b. PWB = 50 ±5 ns.
   c. $t_{TLH1} = t_{THL1} = 3 \pm 1$ ns.
3. Minimum setup time clock:
   a. For test conditions AA1-AA14:
      $t_{SETUP(1)} = 6$ +1, -0 ns, $t_{SETUP(0)} = 12$ +2, -0 ns.
   b. For test conditions AA15-AA28:
      $t_{SETUP(1)} = 6$ +1, -0 ns, $t_{SETUP(0)} = 6$ +1, -0 ns.
   c. The setup time requirements must be verified before making measurements per table XXI.
4. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.5 V before making measurements per table XXI.
5. All voltages are referenced to ground.

FIGURE 32. Setup time test circuit waveforms for table XXI.

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TABLE XXII. Hold time test.

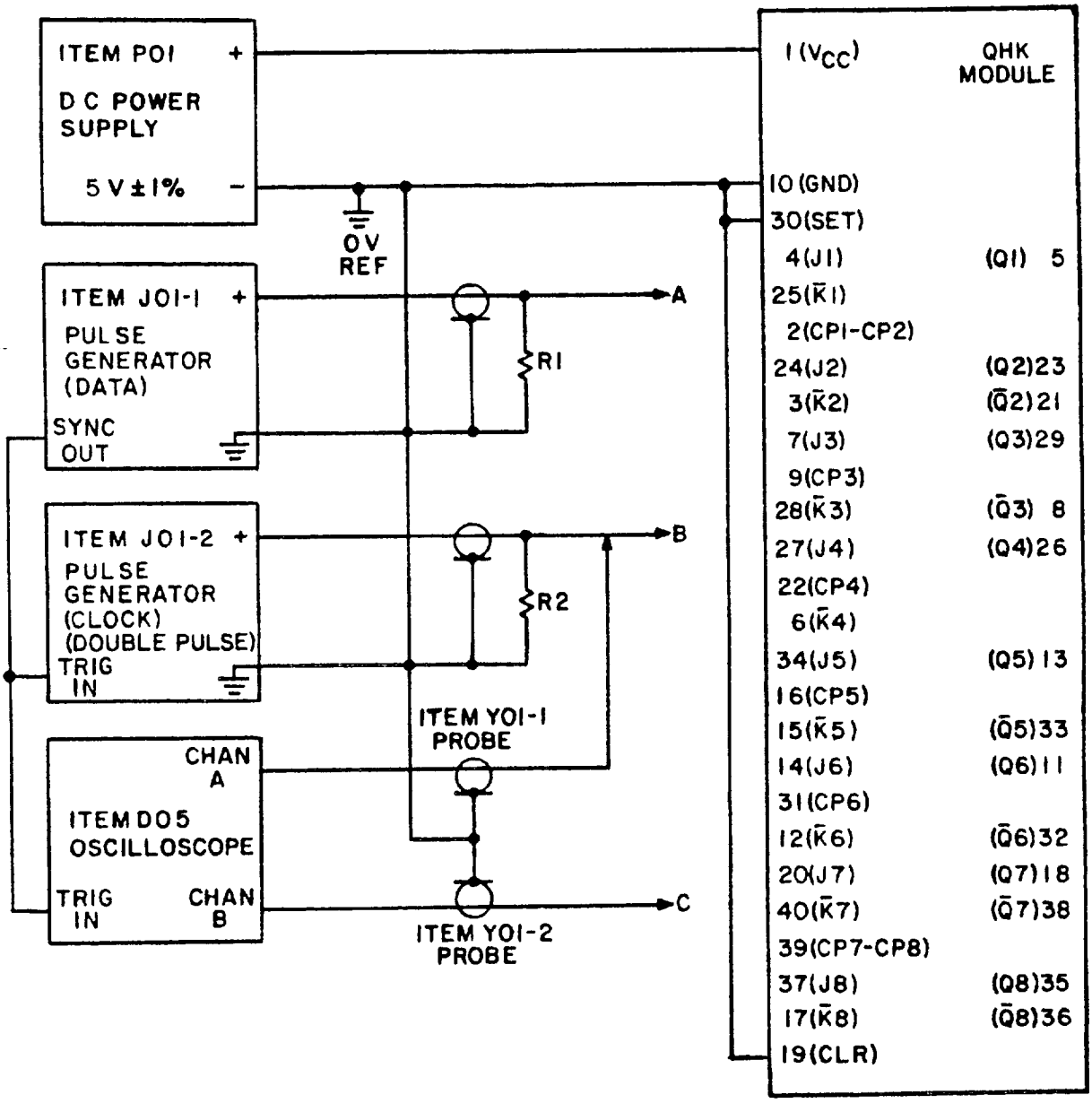
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
BB1	Q1 Setup time, C_L = 50 pF	4,25	2	5	80	120	ns	33,34
BB2	Q2 "	3,24	2	23	"	"	"	"
BB3	Q3 "	7,28	9	29	"	"	"	"
BB4	Q4 "	6,27	22	26	"	"	"	"
BB5	Q5 "	15,34	16	13	"	"	"	"
BB6	Q6 "	12,14	31	11	"	"	"	"
BB7	Q7 "	20,40	39	18	"	"	"	"
BB8	Q8 "	17,37	39	35	"	"	"	"
BB9	$\overline{Q2}$ "	3,24	2	21	"	"	"	"
BB10	$\overline{Q3}$ "	7,28	9	8	"	"	"	"
BB11	$\overline{Q5}$ "	15,34	16	33	"	"	"	"
BB12	$\overline{Q6}$ "	12,14	31	32	"	"	"	"
BB13	$\overline{Q7}$ "	20,40	39	38	"	"	"	"
BB14	$\overline{Q8}$ "	17,37	39	36	"	"	"	"
BB15	Q1 "	4,25	2	5	"	"	"	"
BB16	Q2 "	3,24	2	23	"	"	"	"
BB17	Q3 "	7,28	9	29	"	"	"	"
BB18	Q4 "	6,27	22	26	"	"	"	"
BB19	Q5 "	15,34	16	13	"	"	"	"
BB20	Q6 "	12,14	31	11	"	"	"	"

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TABLE XXII. Hold time test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
BB21	Q7 Setup time, C_L = 50 pF	20,40	39	18	80	120	ns	33,34
BB22	Q8 "	17,37	39	35	"	"	"	"
BB23	$\overline{Q2}$ "	3,24	2	21	"	"	"	"
BB24	$\overline{Q3}$ "	7,28	9	8	"	"	"	"
BB25	$\overline{Q5}$ "	15,34	16	33	"	"	"	"
BB26	$\overline{Q6}$ "	12,14	31	32	"	"	"	"
BB27	$\overline{Q7}$ "	20,40	39	38	"	"	"	"
BB28	$\overline{Q8}$ "	17,37	39	36	"	"	"	"

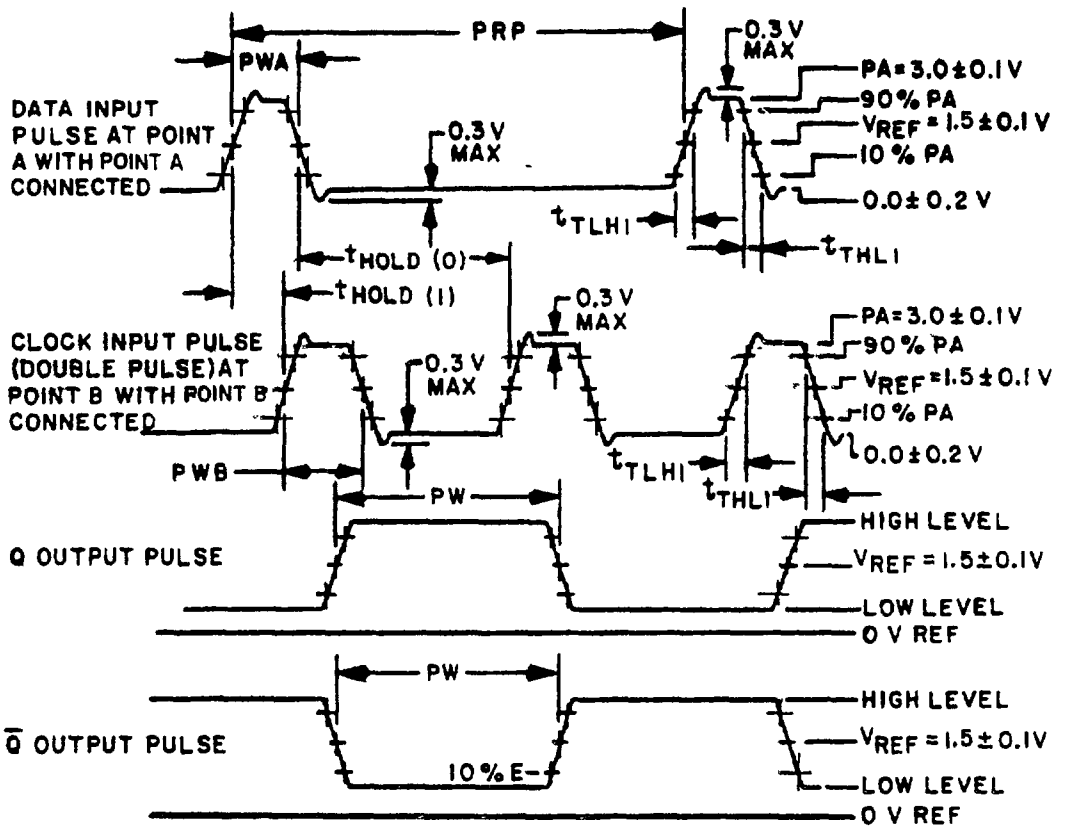
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs as shown on figure 14.

FIGURE 33. Hold time test circuit for table XXII.

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NOTES:
1. Data input pulse parameters:
   a. PRP = 200 ±20 ns.
   b. $t_{TLH1} = t_{THL1} = 3 \pm 1$ ns.
   c. PWA = 100 ±10 ns.
2. Clock input pulse parameters:
   a. Item 301-2 is operated in double pulse mode.
   b. PWB = 50 ±5 ns.
   c. $t_{TLH1} = t_{THL1} = 3 \pm 1$ ns.
3. Minimum setup time clock:
   a. For test conditions BB1-BB14:
      $t_{HOLD(1)} = 0$ +3, -0 ns, $t_{HOLD(0)} = 2$ +2, -0 ns.
   b. For test conditions BB15-BB28:
      $t_{HOLD(1)} = 2$ +2, -0 ns, $t_{HOLD(0)} = 0$ +3, -0 ns.
   c. The hold time requirements must be verified before making measurements per table XXII.
4. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.5 V before making measurements per table XXII.
5. All voltages are referenced to ground.

FIGURE 34. Hold time test circuit waveforms for table XXII.

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